# 第2章 电阻电路的等效变换


2.1 电路的等效变换

2.2 电阻的串联和并联

2.3 电阻的Y形连接和△形连接的等效变换

2.4 电压源、电流源的串联和并联

2.5 实际电源的两种模型及其等效变换

2.6 输入电阻

# 本章要求

- 掌握电阻的Y形连接和△形连接的等效变换；
- 掌握实际电源的两种模型及其等效变换；
- 掌握输入电阻的计算方法。

# 引言

**线性电路：**

由时不变线性无源元件、线性受控源和独立电源组成的电路，称为时不变线性电路，简称线性电路。

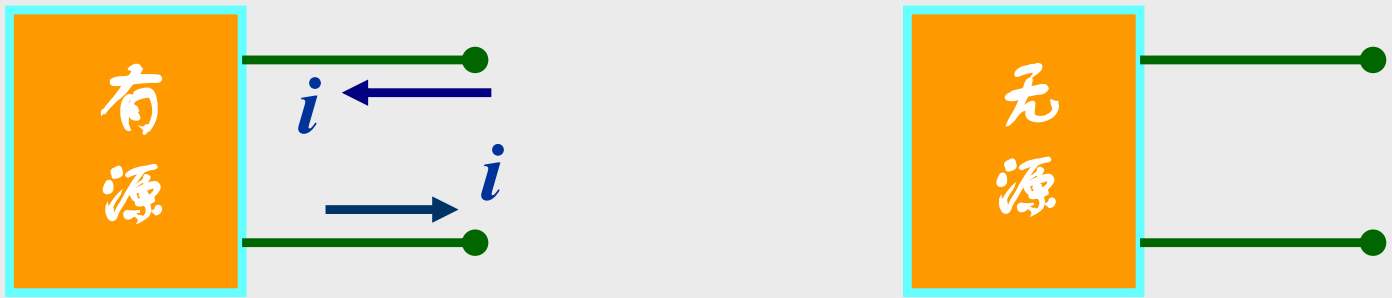
**电阻电路：**

构成电路的无源元件均为线性电阻，则称该电路为线性电阻性电路，简称电阻电路。

# 2.1 电路的等效变换

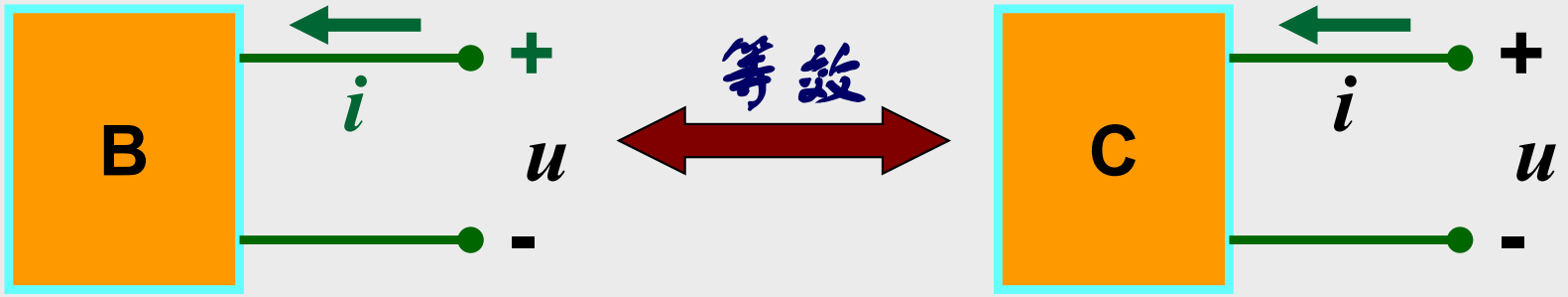
## 两端电路（two-terminal circuit）

任何一个复杂的电路，向外引出两个端子，且从一个端子流入的电流等于从另一端子流出的电流，则称这一电路为二端网络（或一端口网络）。

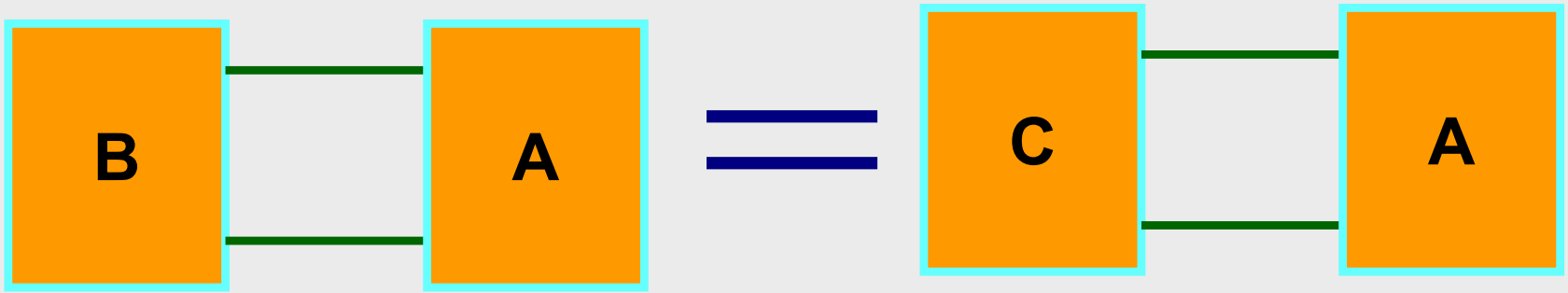



## 两端电路等效的概念

两个两端电路，端口具有相同的电压和电流关系，则称它们是等效的电路。




对A电路中的电流、电压和功率而言，满足

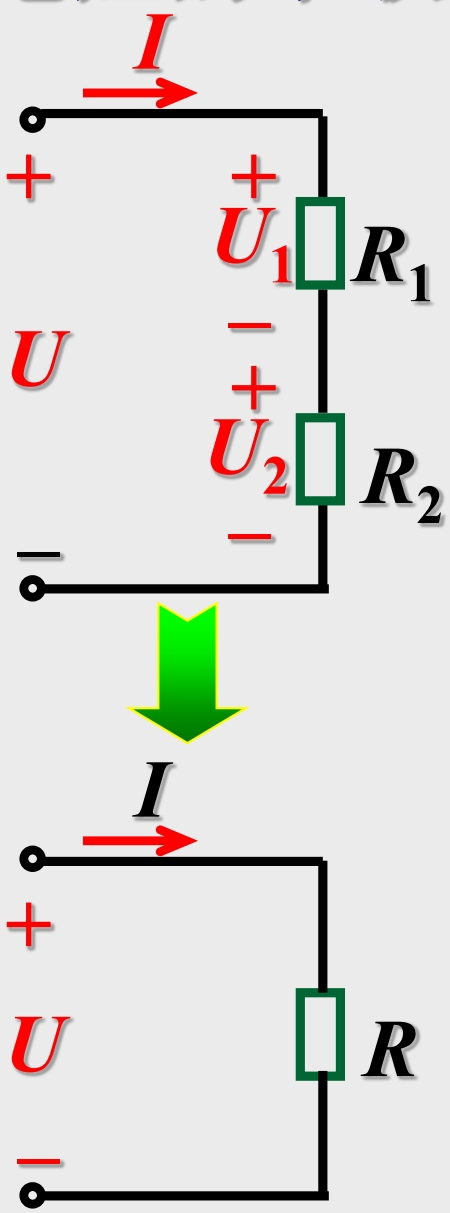



电路等效变换的条件：　两电路具有相同的VCR

电路等效变换的目的：　化简电路，方便计算

# 2.2 电阻的串联和并联

## 电阻的串联




**特点：**

1)各电阻一个接一个地顺序相联；

2)各电阻中通过同一电流；

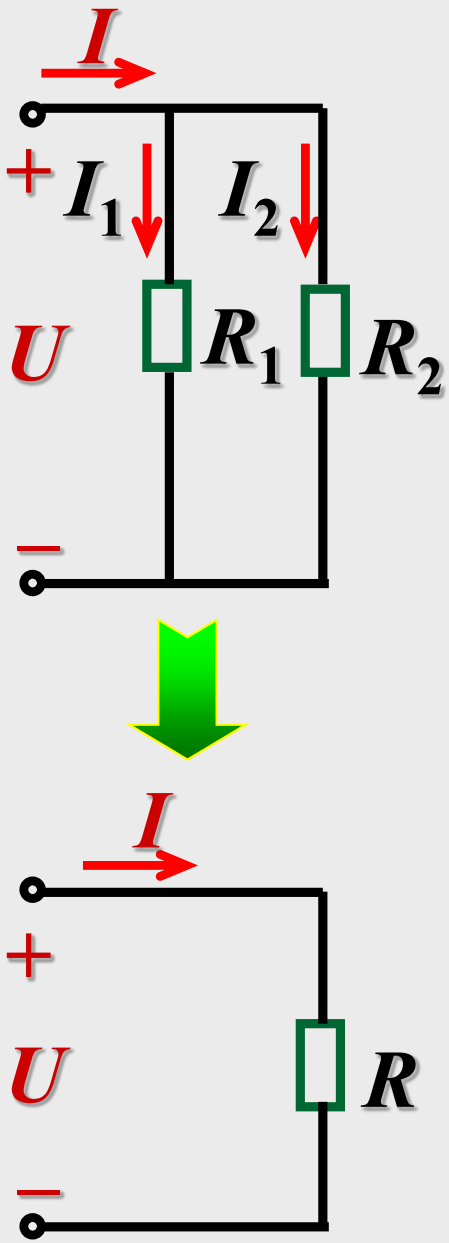
3)等效电阻等于各电阻之和；

$$R = R_1 + R_2$$

$$U = U_1 + U_2 = IR_1 + IR_2 = I(R_1 + R_2)$$

$$U = IR$$

# 电阻的并联




特点：

(1)各电阻联接在两个公共的结点之间；

(2)各电阻两端的电压相同；

(3)等效电阻的倒数等于各电阻倒数之和；

$$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}$$

$$I=I_1+I_2=U/R_1+U/R_2=U(1/R_1+1/R_2)$$

$$I=U(1/R)$$

## 1. 两电阻串联时的分压公式：

$$U_1 = \frac{R_1}{R_1 + R_2} U \qquad U_2 = \frac{R_2}{R_1 + R_2} U$$

串联电阻上电压的分配与电阻成正比。

应用：降压、限流、调节电压等。

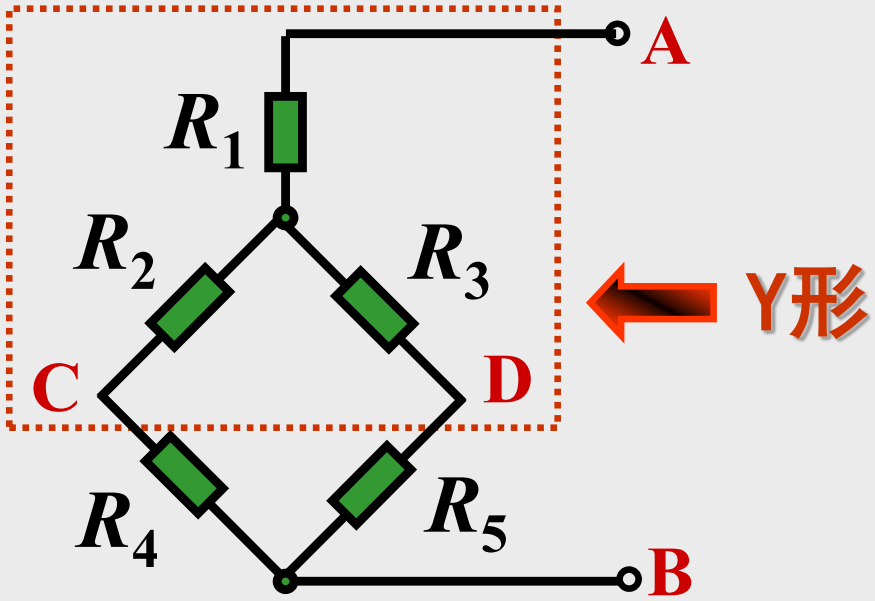
## 2. 两电阻并联时的分流公式：

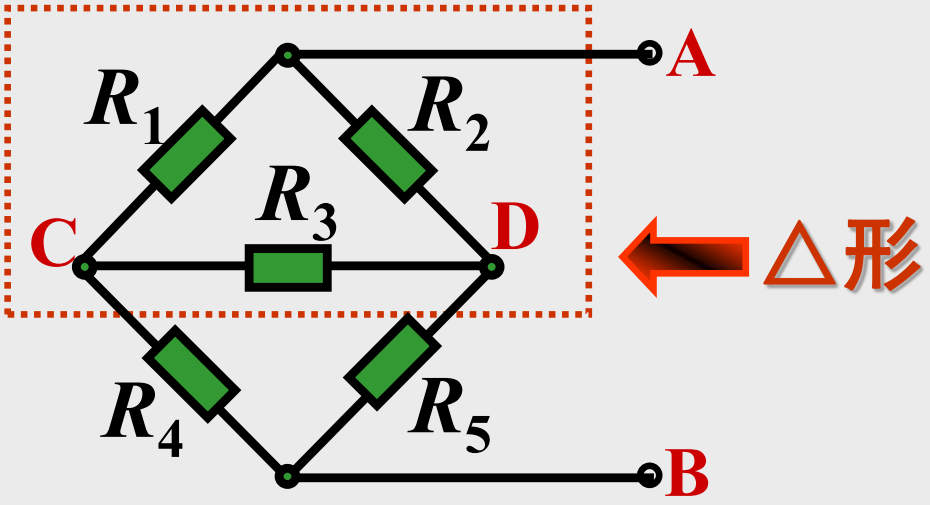
$$I_1 = \frac{R_2}{R_1 + R_2} I \qquad I_2 = \frac{R_1}{R_1 + R_2} I$$

并联电阻上电流的分配与电阻成反比。

应用：分流、调节电流等。

# 2.3 电阻的Y形连接和△形连接的等效变换



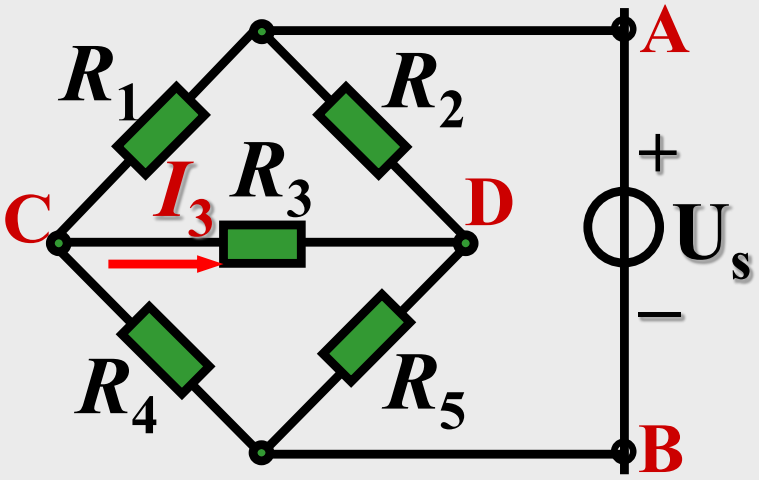




$R = R_1 + [(R_2 + R_4) // (R_3 + R_5)]$

电阻的混联

$R = ?$

电阻的桥形结构




$V_C = V_D$

或：$I_3 = 0$

电桥平衡

惠斯通电桥

$$\frac{U_{AC}}{R_1} = \frac{U_{CB}}{R_4} \qquad \frac{U_{AD}}{R_2} = \frac{U_{DB}}{R_5}$$

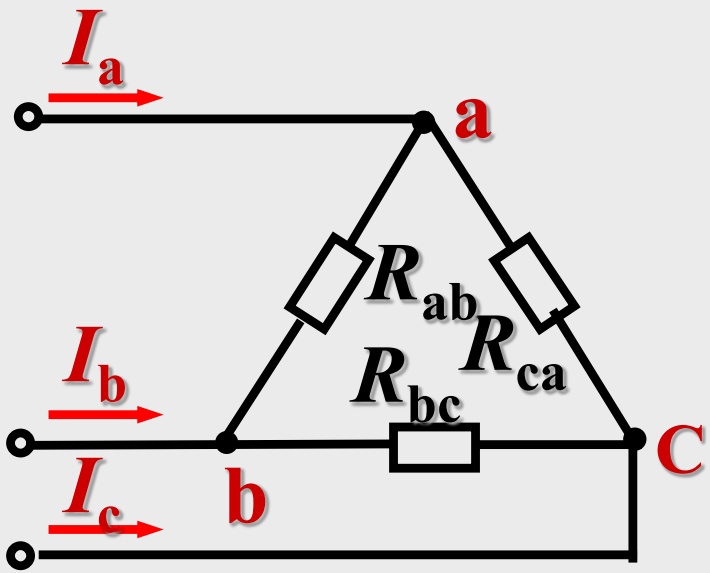
$$\frac{U_{AC}}{R_1} = \frac{U_{CB}}{R_4} \qquad \frac{U_{AC}}{R_2} = \frac{U_{DC}}{R_5}$$

$$\frac{R_2}{R_1} = \frac{R_5}{R_4}$$

$$R_2 R_4 = R_1 R_5$$

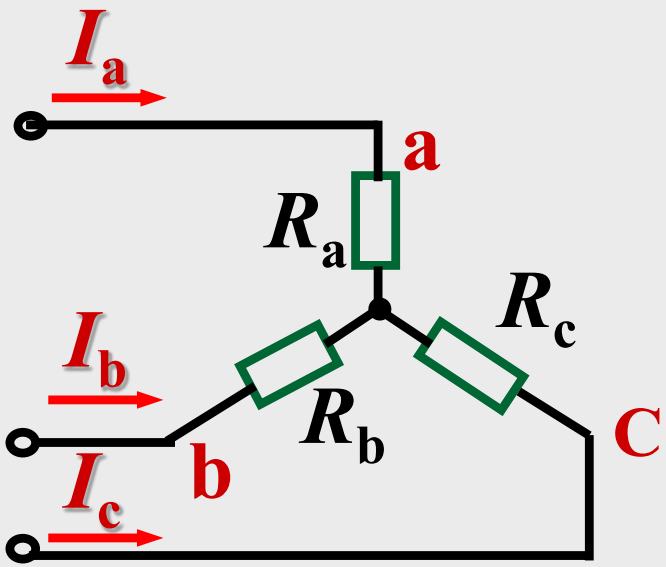
$$R = (R_1 + R_4) // (R_2 + R_5)$$

# 电桥不平衡时：




电阻Δ形连接

等效变换



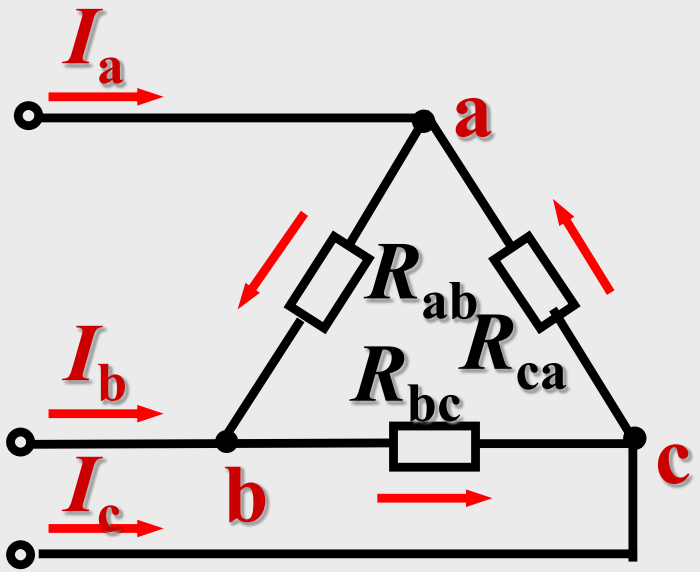

电阻Y形连接

等效变换的条件：

对应端流入或流出的电流（$I_a$、$I_b$、$I_c$）一一相等，对应端间的电压（$U_{ab}$、$U_{bc}$、$U_{ca}$）也一一相等。

经等效变换后，不影响其它部分的电压和电流。

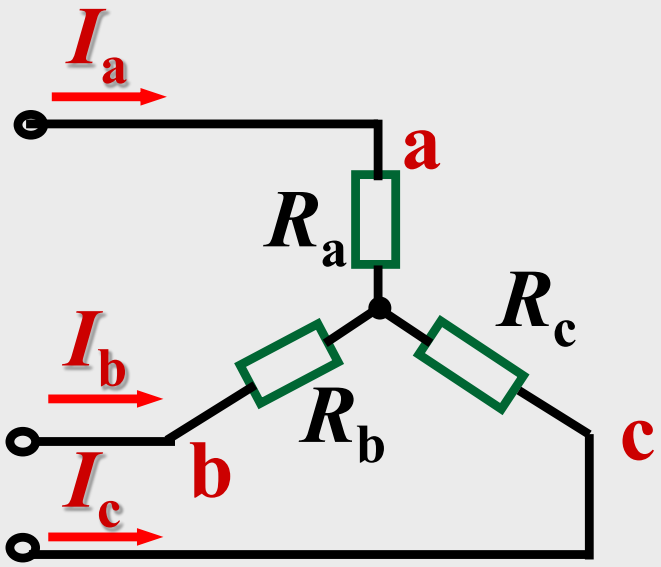
# 电桥不平衡时：




电阻Δ形连接

用电压表示电流

等效变换




电阻Y形连接

用电流表示电压

$$\left.\begin{aligned} I_a &= U_{ab}/R_{ab} - U_{ca}/R_{ca} \\ I_b &= U_{bc}/R_{bc} - U_{ab}/R_{ab} \\ I_c &= U_{ca}/R_{ca} - U_{bc}/R_{bc} \end{aligned}\right\} \quad (1)$$

$$\left.\begin{aligned} U_{ab} &= R_a I_a - R_b I_b \\ U_{bc} &= R_b I_b - R_c I_c \\ U_{ca} &= R_c I_c - R_a I_a \\ I_a + I_b + I_c &= 0 \end{aligned}\right\} \quad (2)$$

由式(2)解得：

$$I_a = \frac{U_{ab}R_c - U_{ca}R_b}{R_aR_b + R_bR_c + R_cR_a}$$
$$I_b = \frac{U_{bc}R_a - U_{ab}R_c}{R_aR_b + R_bR_c + R_cR_a}$$
$$I_c = \frac{U_{ca}R_b - U_{bc}R_a}{R_aR_b + R_bR_c + R_cR_a}$$
(3)

$$I_a = U_{ab}/R_{ab} - U_{ca}/R_{ca}$$
$$I_b = U_{bc}/R_{bc} - U_{ab}/R_{ab}$$
$$I_c = U_{ca}/R_{ca} - U_{bc}/R_{bc}$$
(1)

根据等效条件，比较式(3)与式(1)，得Y型→Δ型的变换条件：

$$\frac{1}{R_{ab}} = \frac{R_c}{R_aR_b + R_bR_c + R_cR_a}$$
$$\frac{1}{R_{bc}} = \frac{R_a}{R_aR_b + R_bR_c + R_cR_a}$$
$$\frac{1}{R_{ca}} = \frac{R_b}{R_aR_b + R_bR_c + R_cR_a}$$

或

$$G_{ab} = \frac{G_aG_b}{G_a + G_b + G_c}$$
$$G_{bc} = \frac{G_bG_c}{G_a + G_b + G_c}$$
$$G_{ca} = \frac{G_cG_a}{G_a + G_b + G_c}$$

类似可得到由Δ型→ Y型的变换条件：

$$G_a = G_{ab} + G_{ca} + \frac{G_{ab}G_{ca}}{G_{bc}}$$

$$G_b = G_{bc} + G_{ab} + \frac{G_{bc}G_{ab}}{G_{ca}}$$

$$G_c = G_{ca} + G_{bc} + \frac{G_{ca}G_{bc}}{G_{ab}}$$

或

$$R_a = \frac{R_{ab}R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

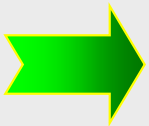
$$R_b = \frac{R_{bc}R_{ab}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{ca}R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

**当Y形连接和△形连接的三个电阻相等时：**

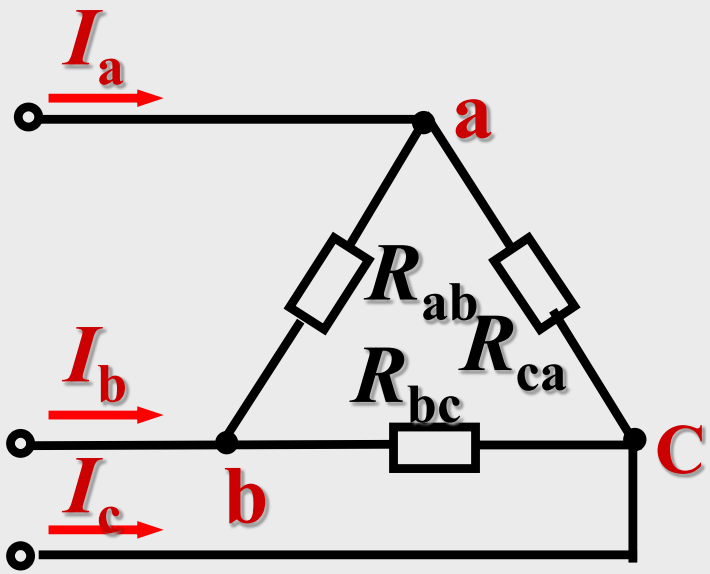
$$R_a = R_b = R_c = R_Y$$

$$R_{ab} = R_{bc} = R_{ca} = R_\Delta$$



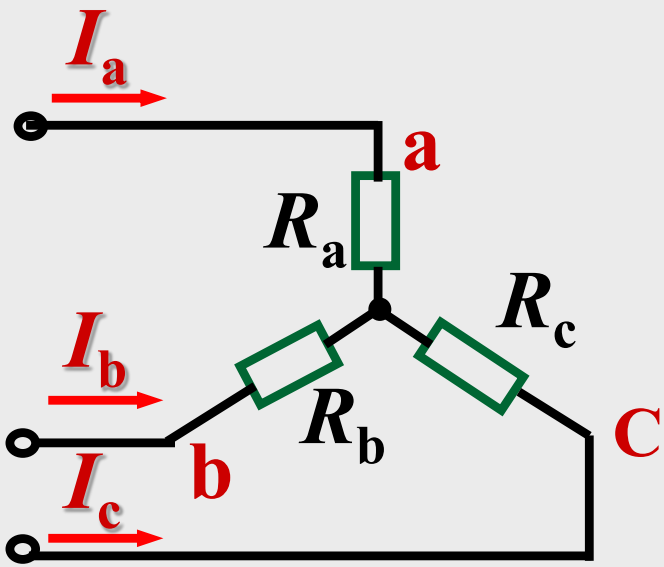
$$R_Y = \frac{1}{3} R_\Delta$$

$$R_\Delta = 3R_Y$$




电阻Δ形连接

等效变换




电阻Y形连接

**简记方法：**

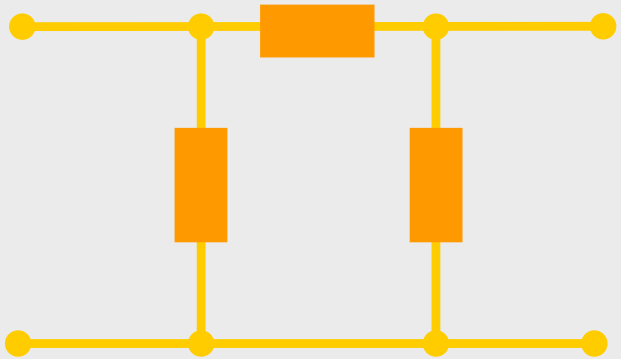
$$R_Y = \frac{\Delta\text{相邻电阻乘积}}{\sum R_\Delta}$$

Δ变Y

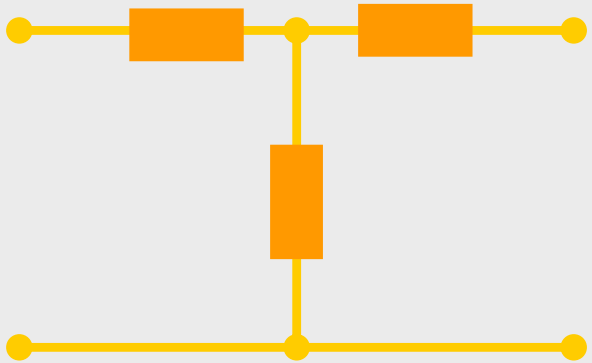
$$G_\Delta = \frac{\text{Y相邻电导乘积}}{\sum G_Y}$$

Y变Δ

# Δ，Y 网络的变形：



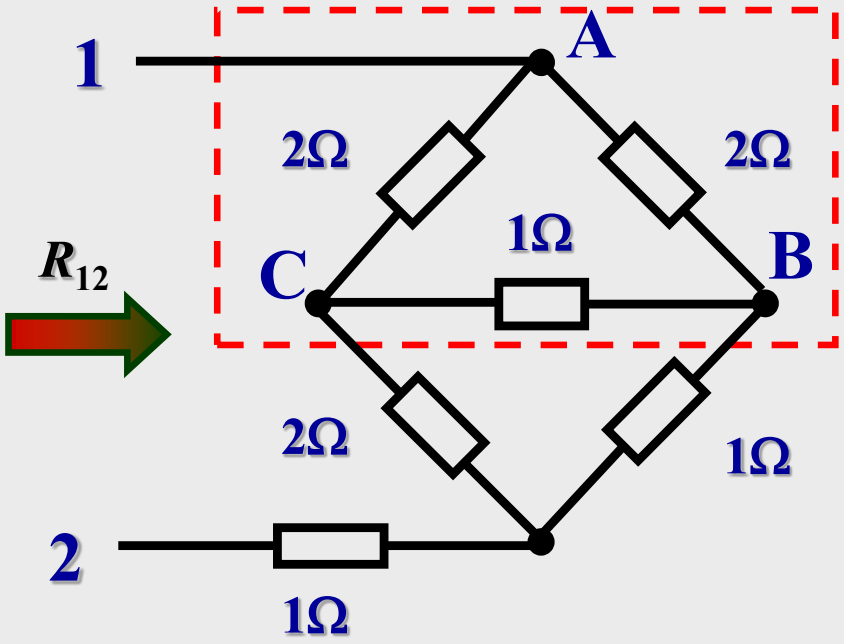
π 型电路（Δ 型）

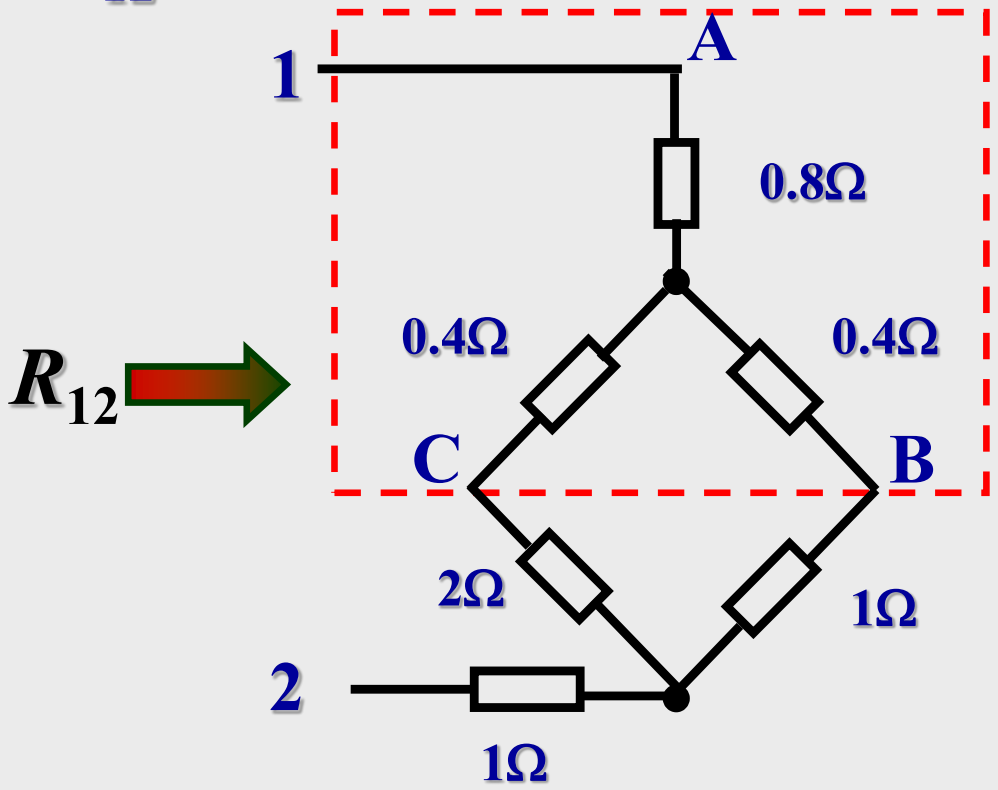


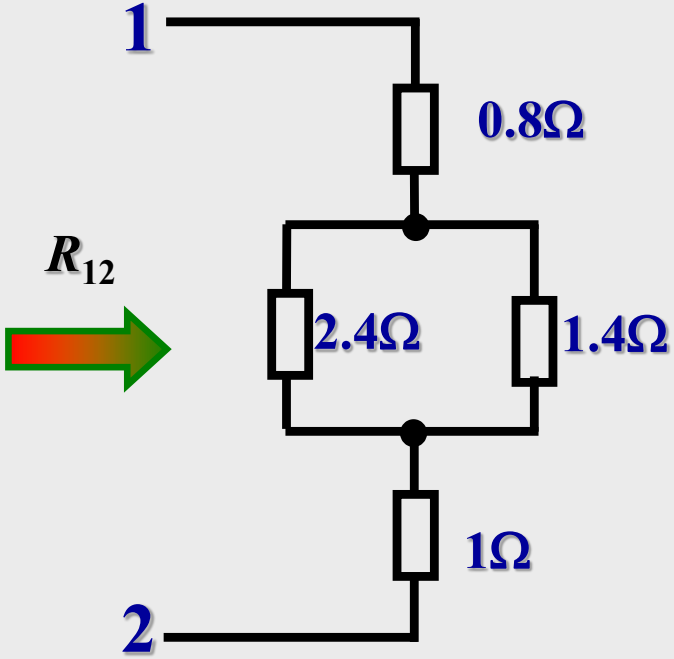
T 型电路（Y型）

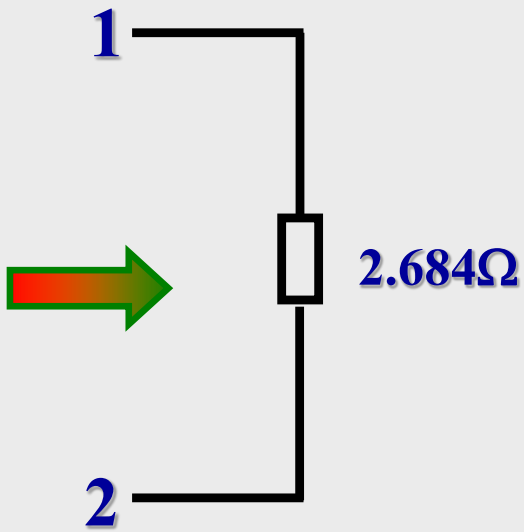
这两个电路当它们的电阻满足一定的关系时，能够相互等效

# 例1：对图示电路求总电阻$R_{12}$

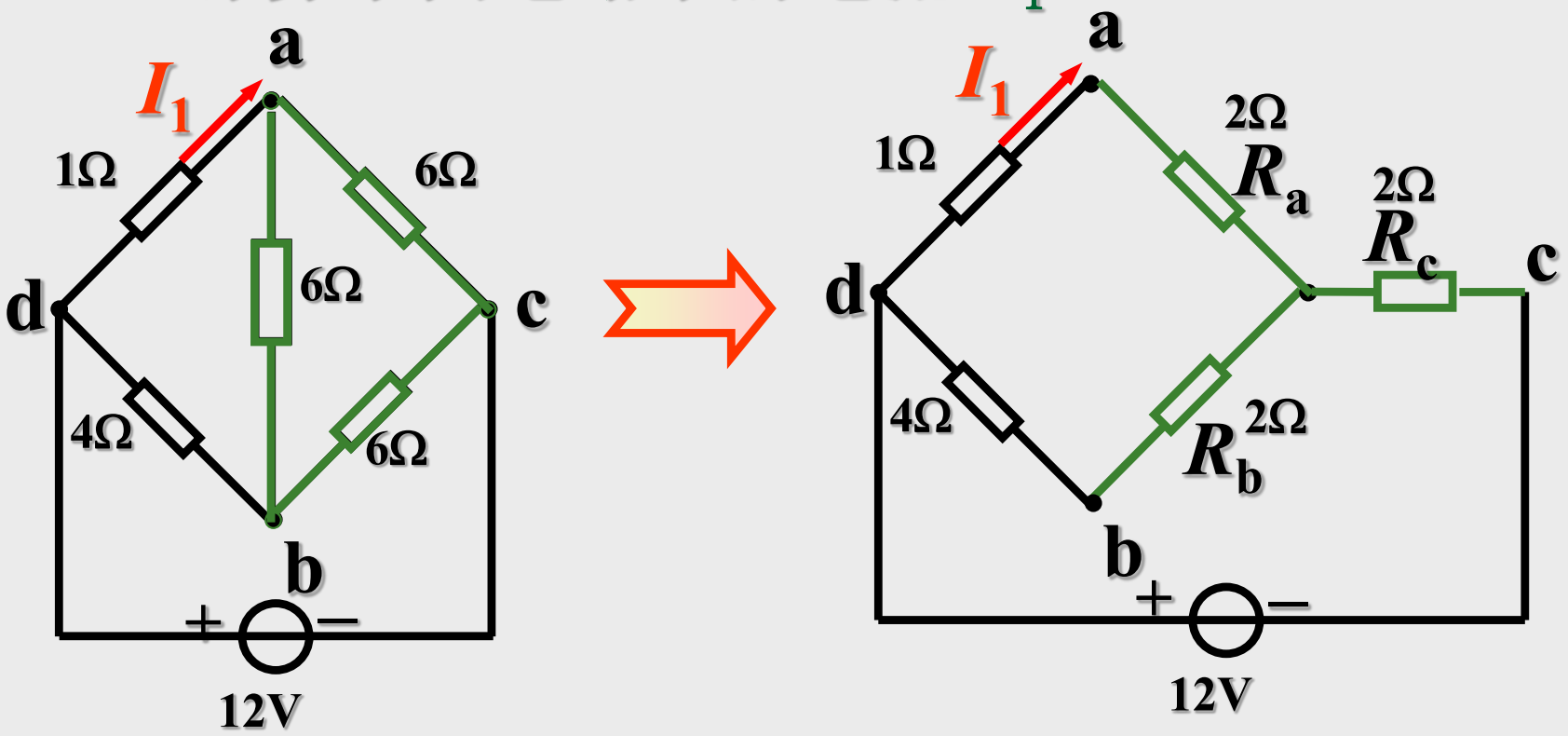


例2：计算下图电路中的电流 $I_1$ 。

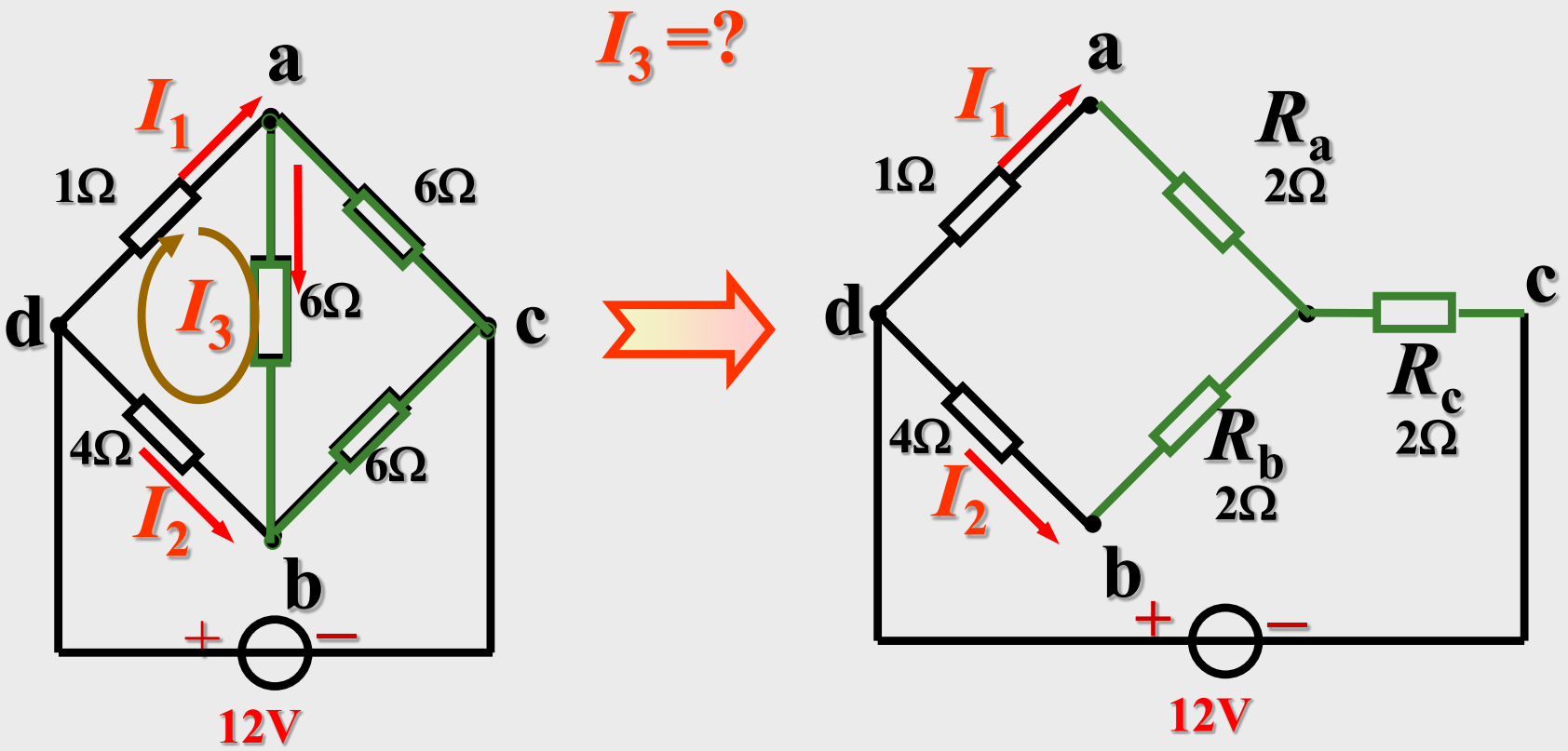



解：将联成Δ形abc的电阻变换为Y形连接的等效电阻

$$R_a = R_b = R_c = 2\ \Omega$$

$$R = \frac{3*6}{3+6} + 2 = 4\ \Omega$$

$$I_1 = \frac{12}{4} * \frac{4+2}{1+2+4+2} = 2\text{A}$$



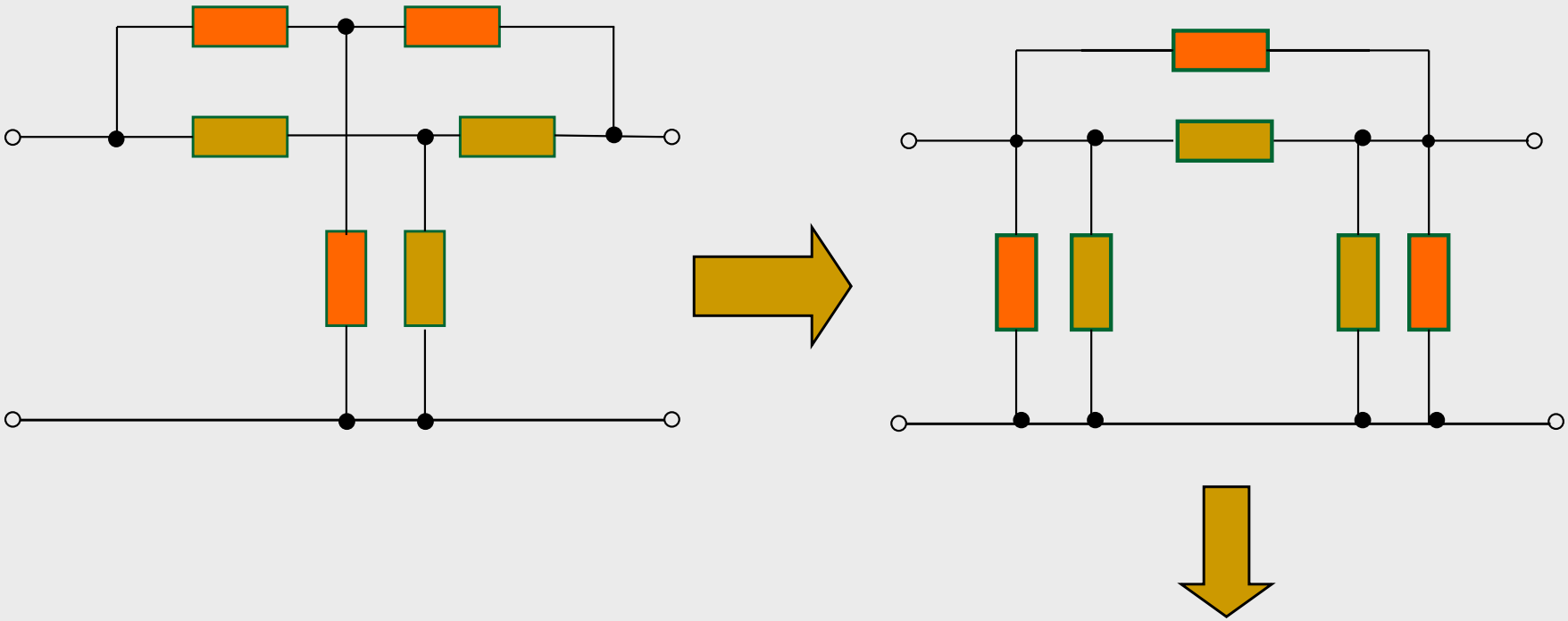

$$I_1 = \frac{12}{4} * \frac{4+2}{1+2+4+2} = 2A$$

$$I_2 = \frac{12}{4} * \frac{1+2}{1+2+4+2} = 1A$$

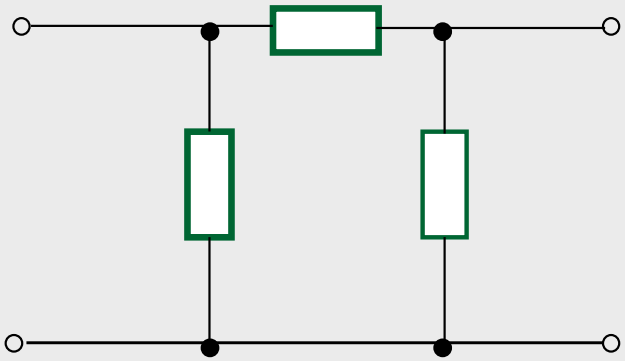
$$I_1 * 1 + I_3 * 6 - I_2 * 4 = 0$$

$$I_3 = \frac{1}{3}A$$

# 例3. 双 T 网络



Y型→Δ型的变换

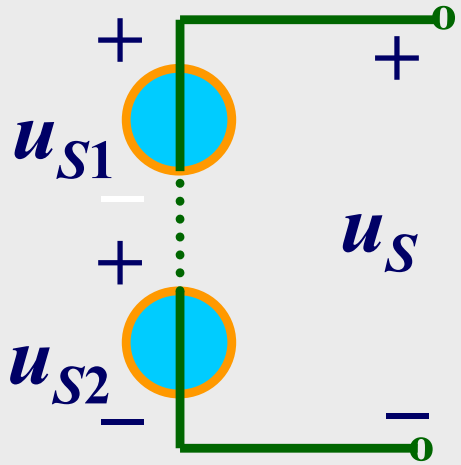


# 2.4 电压源和电流源的串联和并联

## 1. 理想电压源的串联和并联

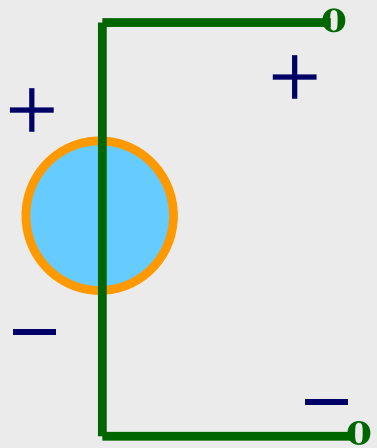
注意参考方向

串联

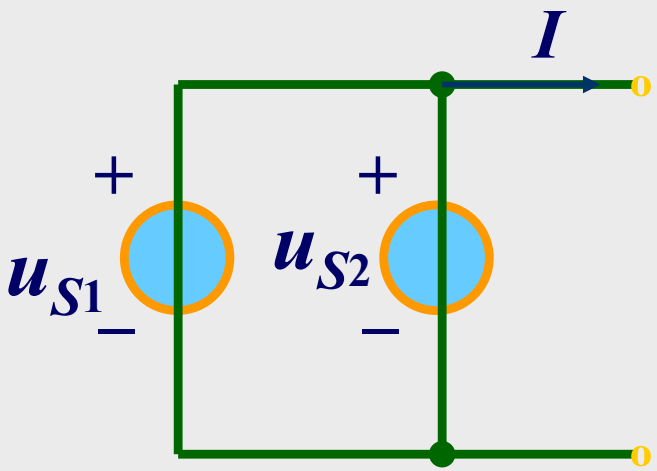



$$u_s = u_{s1} + u_{s2} = \sum u_{sk}$$

等效电路




并联

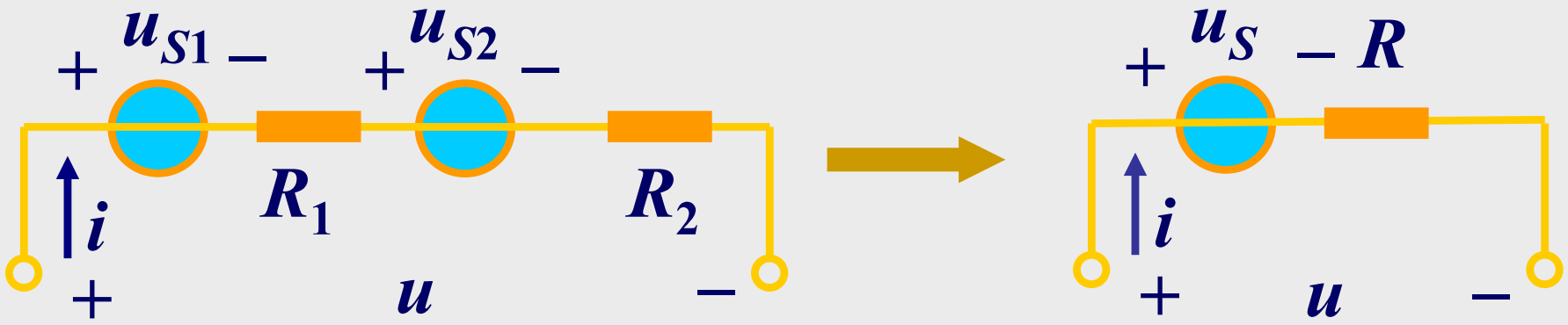



等效电路

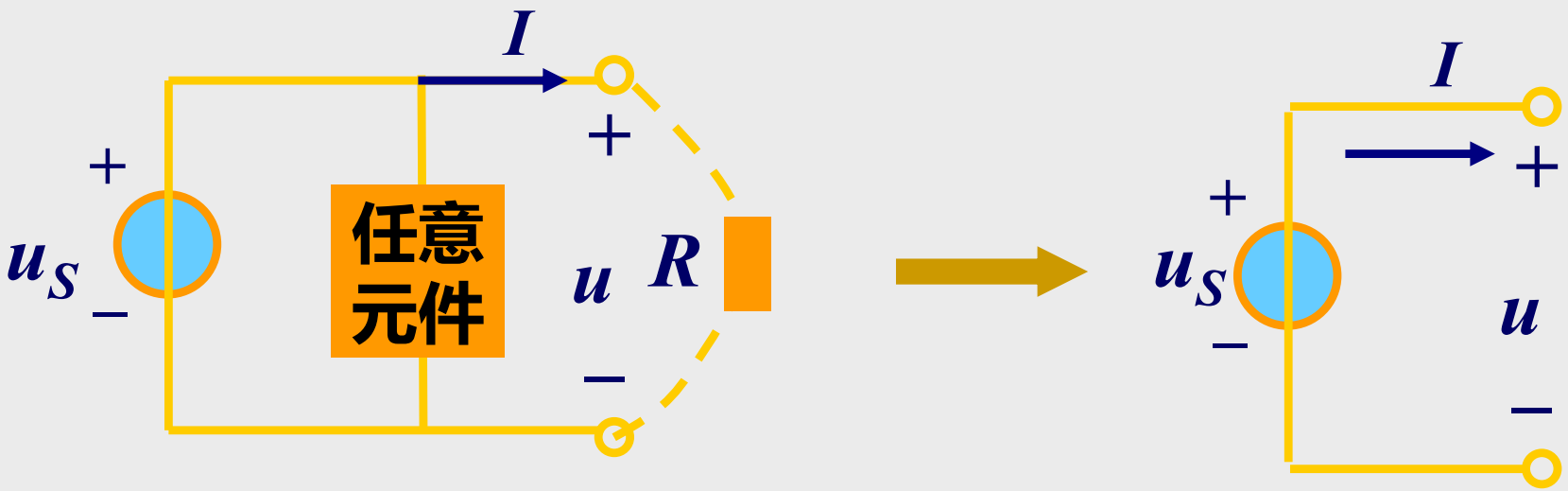
$$u_s = u_{s1} = u_{s2}$$

相同的电压源才能并联，电源中的电流不确定。

# 电压源与支路的串、并联等效




$$u = u_{s1} + R_1 i + u_{s2} + R_2 i = (u_{S1} + u_{S2}) + (R_1 + R_2) i = u_S + Ri$$



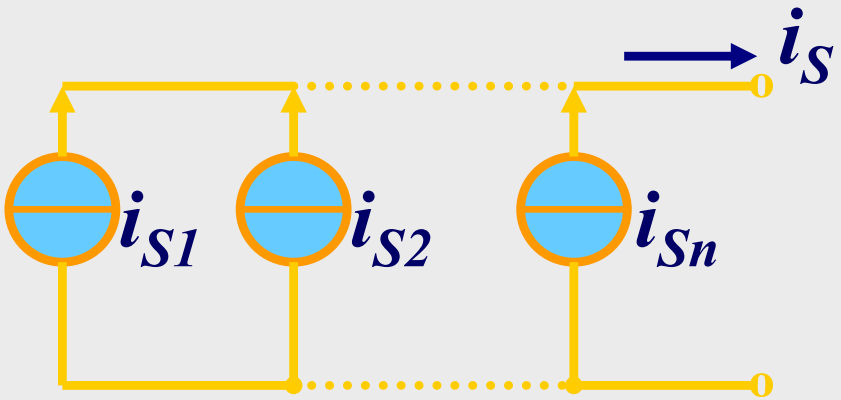

对外等效！

## 2. 理想电流源的串联并联

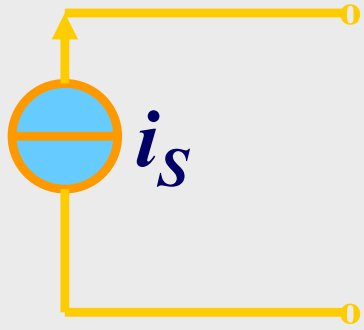
注意参考方向

**并联**

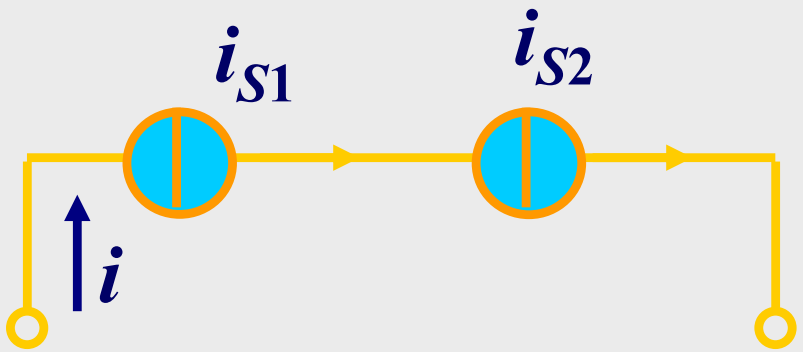
$$i_s = i_{s1} + i_{s2} + \cdots + i_{sn} = \sum i_{sk}$$




等效电路




**串联**

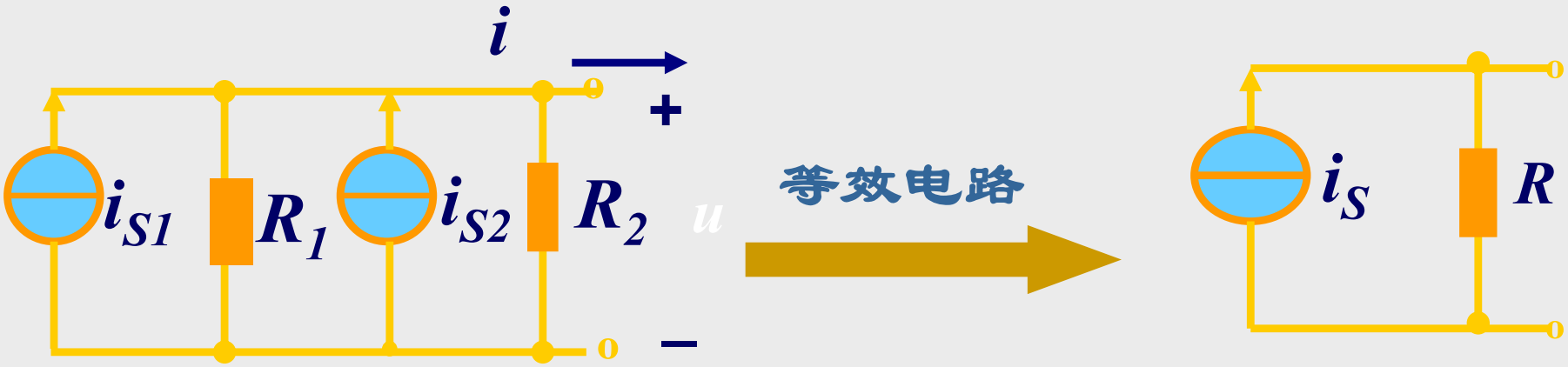



等效电路

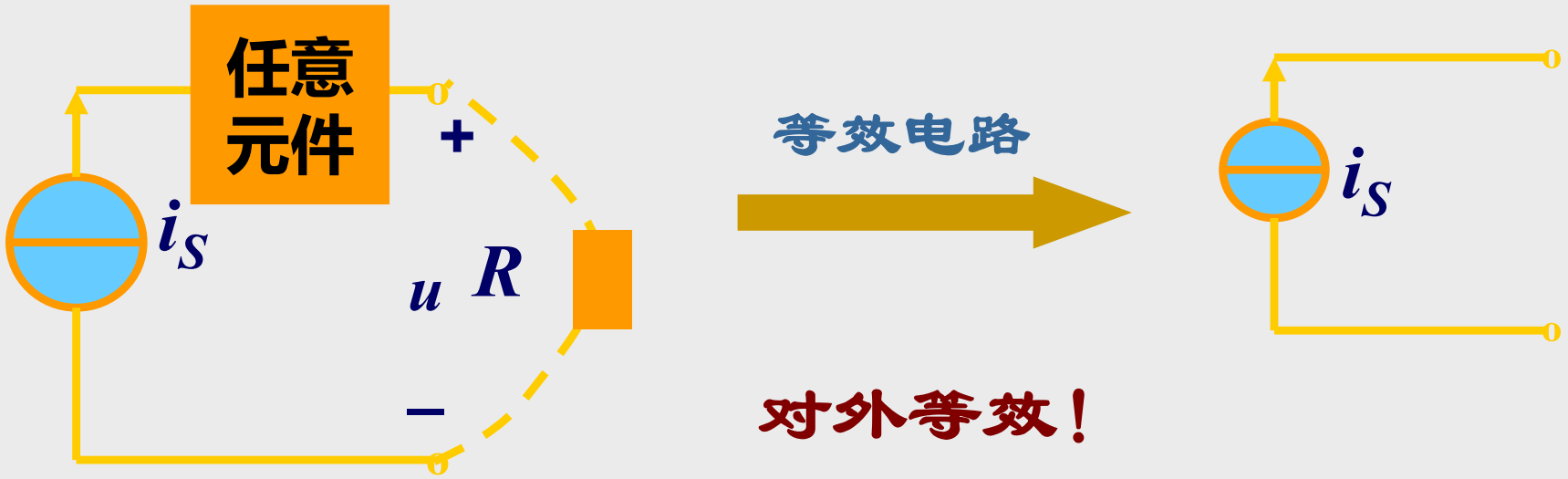
$$i_s = i_{s1} = i_{s2}$$

相同的理想电流源才能串联，每个电流源的端电压不能确定

# 电流源与支路的串、并联等效



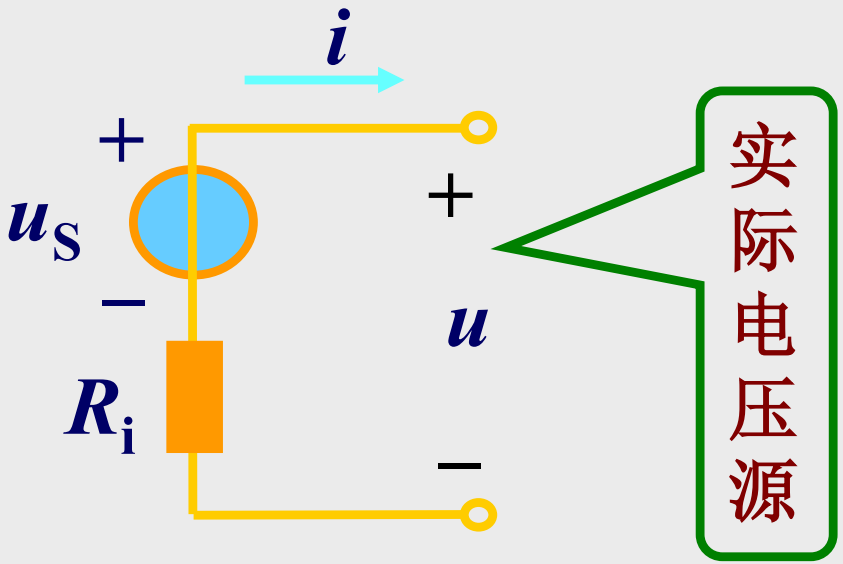

$$i = i_{s1} + u/R_1 + i_{s2} + u/R_2 = i_{s1} + i_{s2} + (1/R_1 + 1/R_2)u = i_s + u/R$$

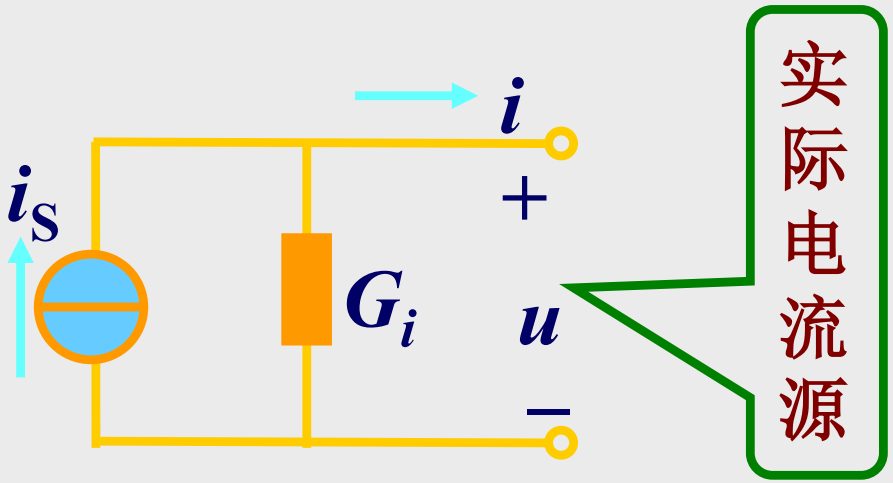



对外等效！

# 2.5 电压源和电流源的等效变换

实际电压源、实际电流源两种模型可以进行等效变换，所谓的等效是指端口的电压、电流在转换过程中保持不变。




端口特性

$$u = u_S - R_i\, i$$

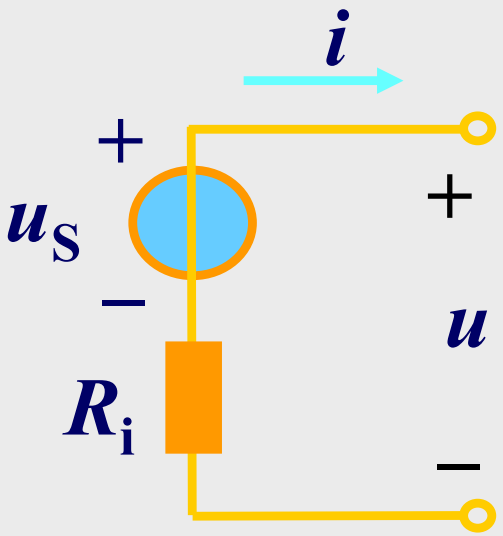
$$i = u_S/R_i - u/R_i$$

$$i = i_S - G_i u$$

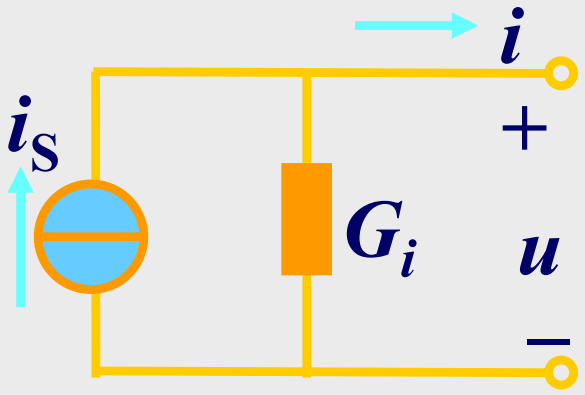
比较可得等效的条件：

$$i_S = u_S/R_i$$
$$G_i = 1/R_i$$

由电压源变换为电流源：

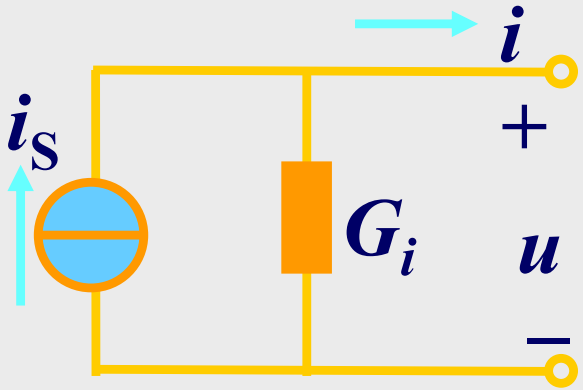



转换

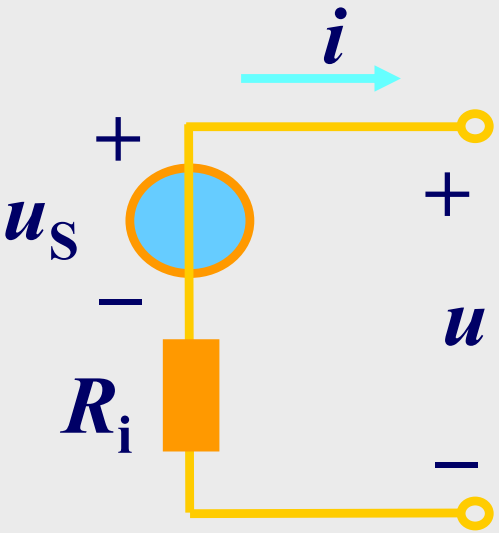



$$i_s = u_s / R_i, \quad G_i = 1 / R_i$$

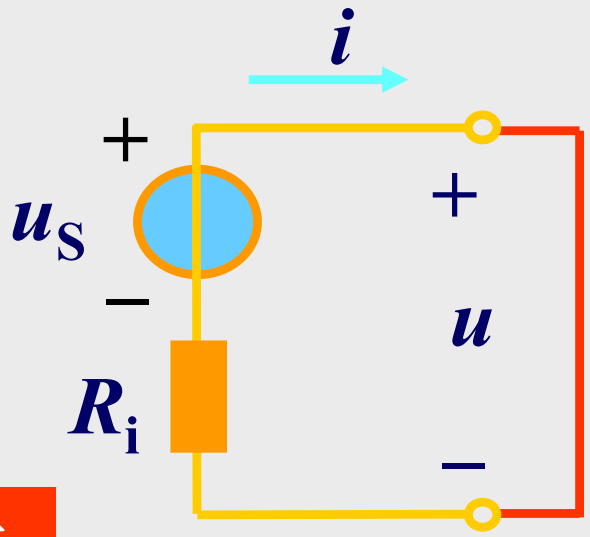
由电流源变换为电压源：

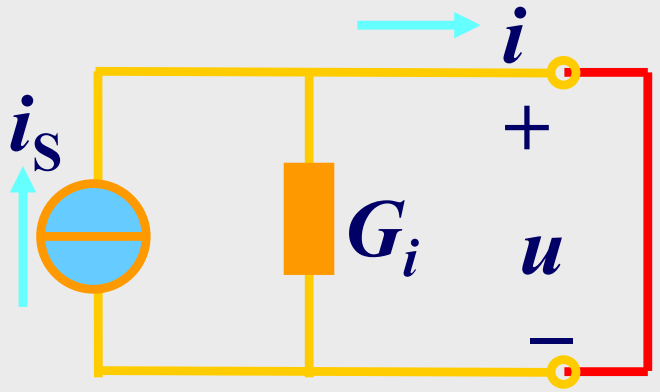



转换




$$u_s = i_s / G_i, \quad R_i = 1 / G_i$$




**注意**

变换关系 [ 数值关系：

方向：电流源电流方向与电压源电压方向相反。

等效是对外部电路等效，对内部电路是不等效的。

**表现在**

开路的电压源中无电流流过 $R_i$；

开路的电流源可以有电流流过并联电导 $G_i$。

电压源短路时，电阻中 $R_i$ 有电流；

电流源短路时，并联电导 $G_i$ 中无电流。

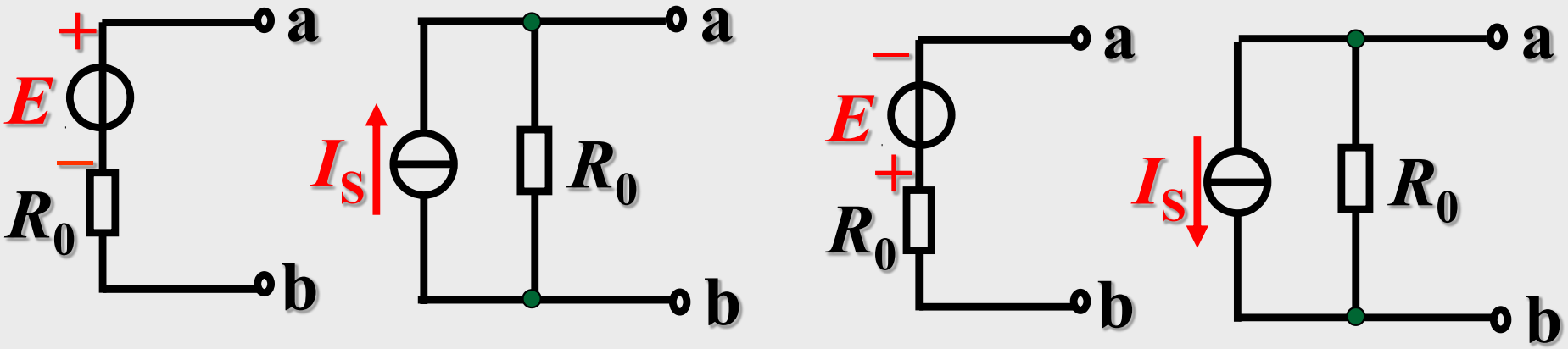
理想电压源与理想电流源不能相互转换。

## 注意事项：

① 电压源和电流源的等效关系只对外电路而言，对电源内部则是不等效的。

例：当$R_L=\infty$时，电压源的内阻$R_0$中不损耗功率，而电流源的内阻$R_0$中则损耗功率。

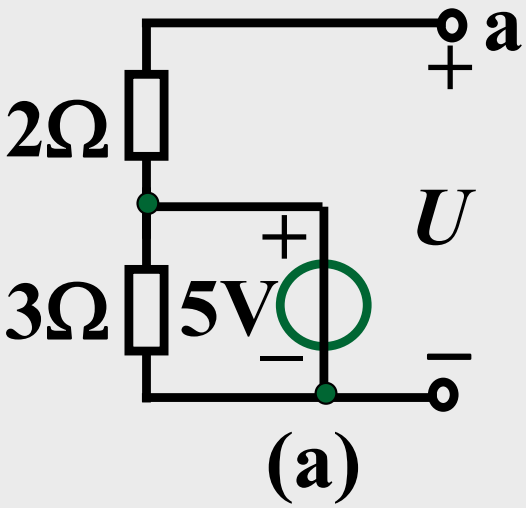
② 等效变换时，两电源的参考方向要一一对应。



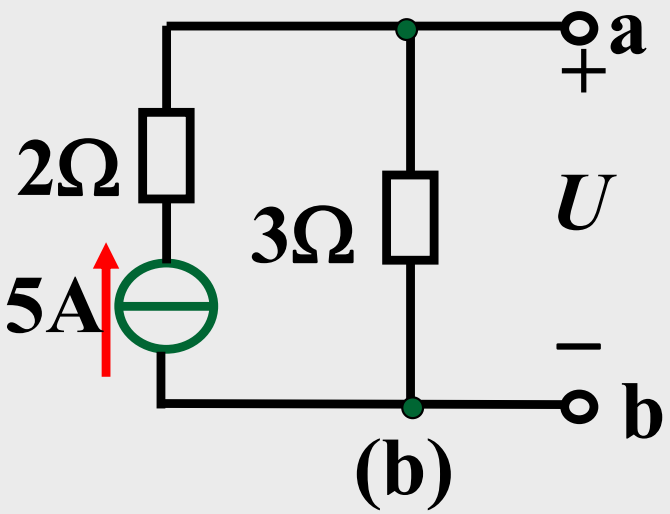

③ 理想电压源与理想电流源之间无等效关系。

④ 任何一个电动势$E$和某个电阻$R$串联的电路，都可化为一个电流为$I_S$和这个电阻并联的电路。

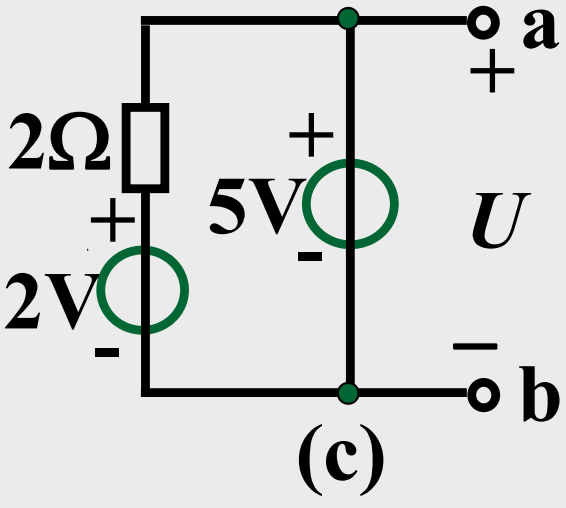
例: 求下列各电路的等效电源



(a)

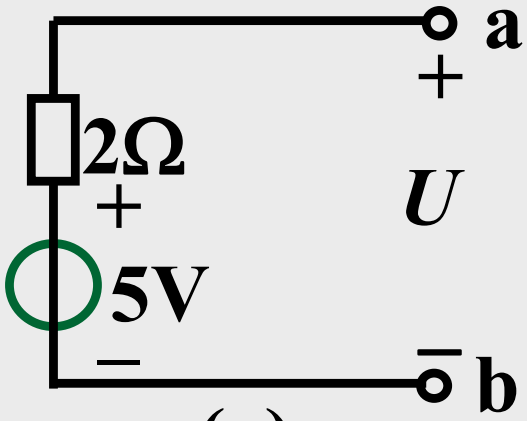


(b)

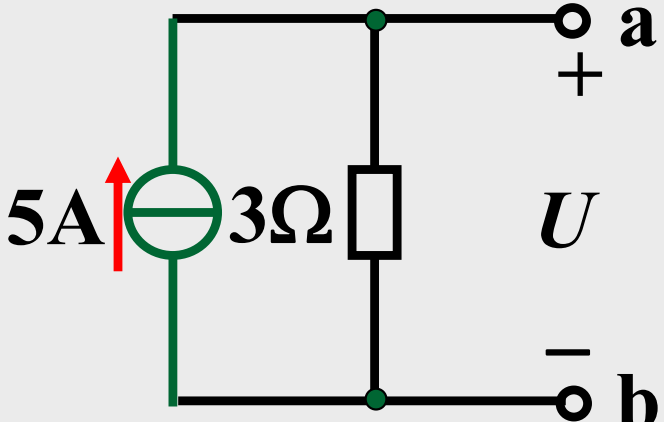


(c)

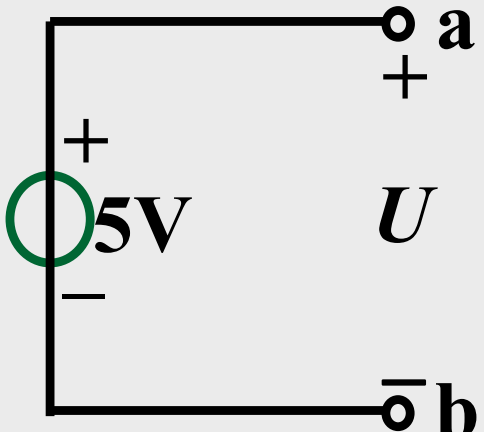
解:



(a)

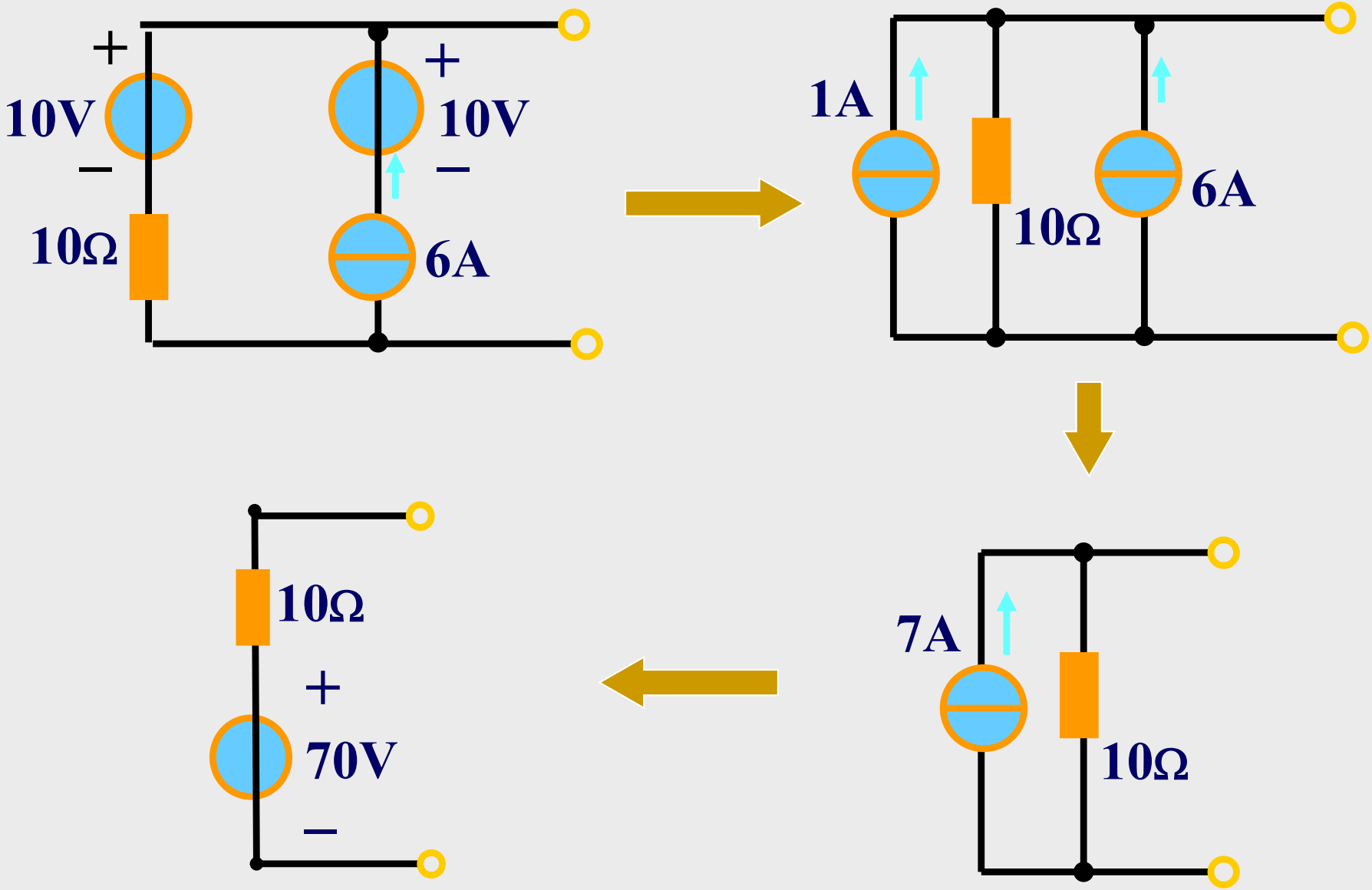


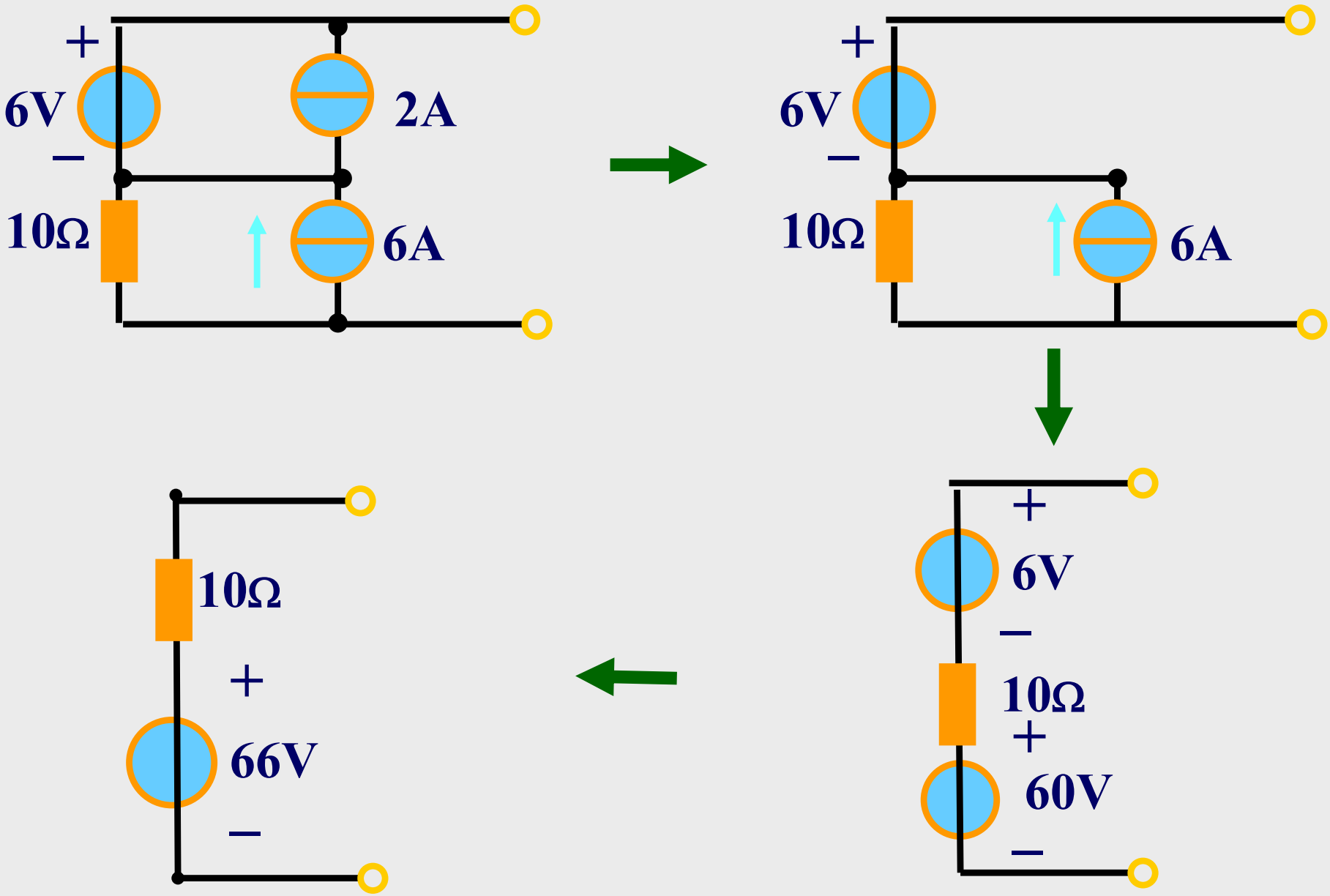
(b)



(c)

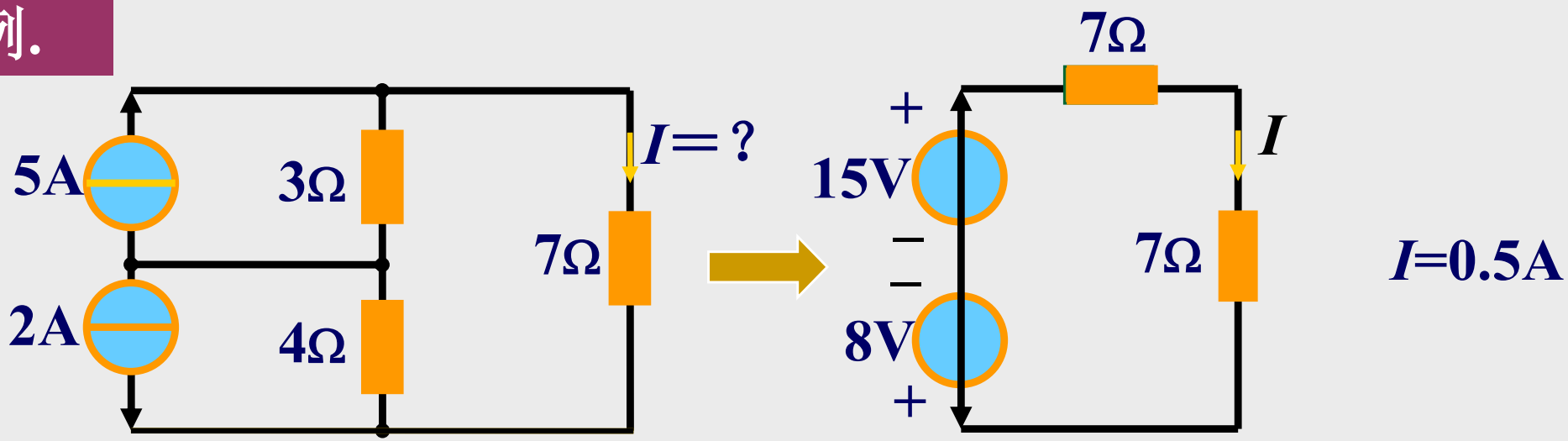
**例.** 把电路转换成一个电压源和一个电阻的串连。




+
6V
−
10Ω
2A
6A
+
6V
−
10Ω
6A
+
6V
−
10Ω
+
60V
−
10Ω
+
66V
−

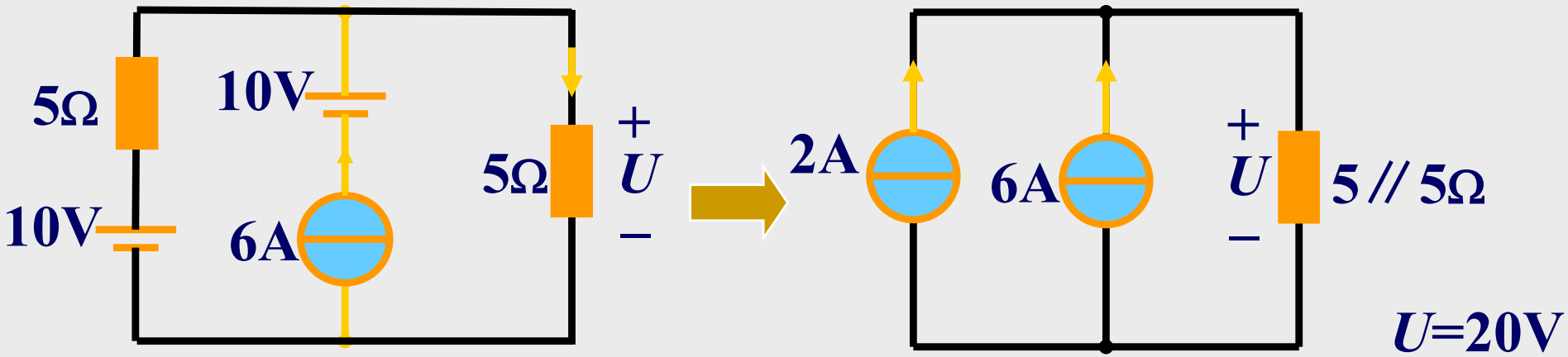
利用电源转换简化电路计算。

例.

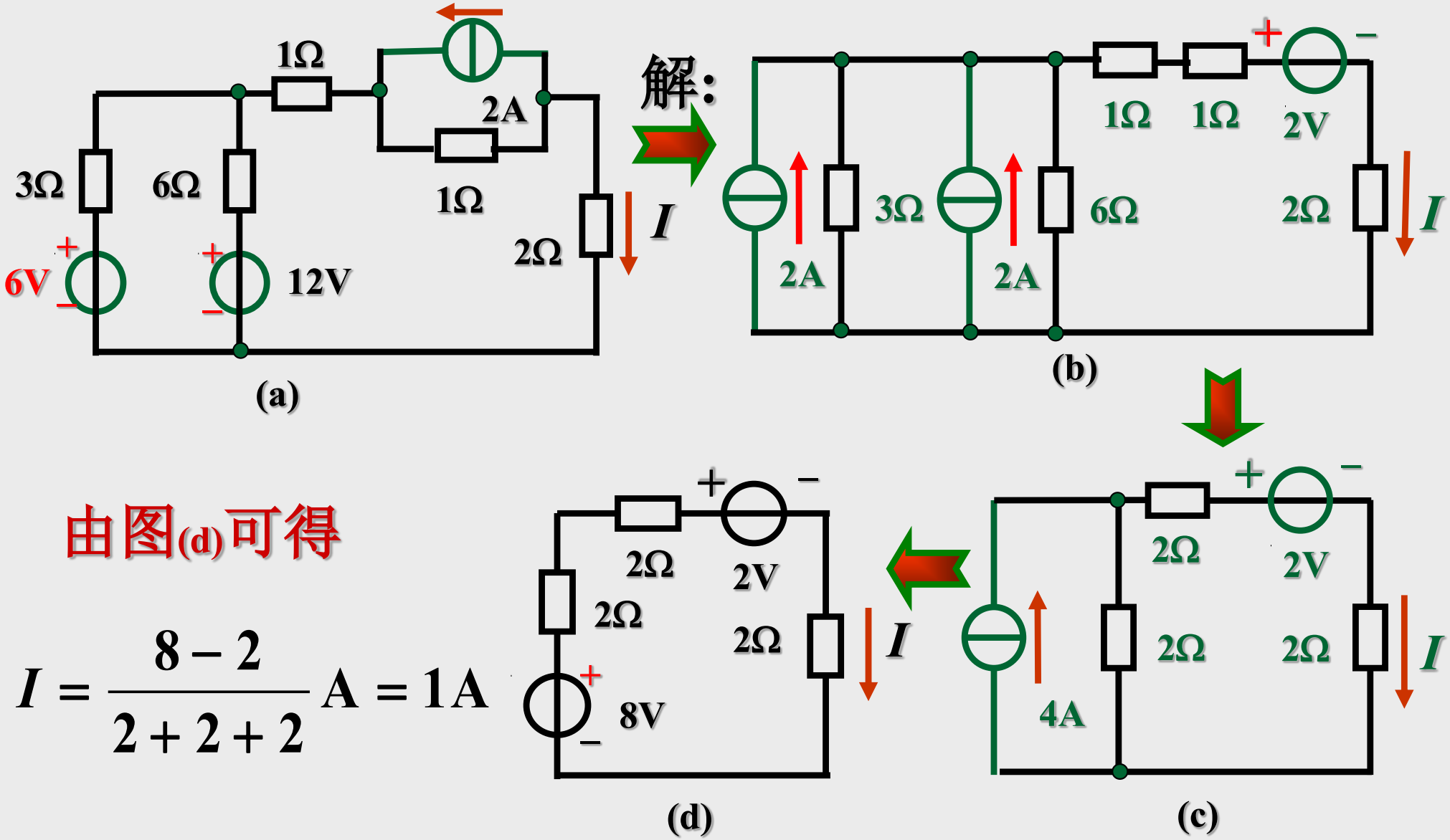



例.

U=?



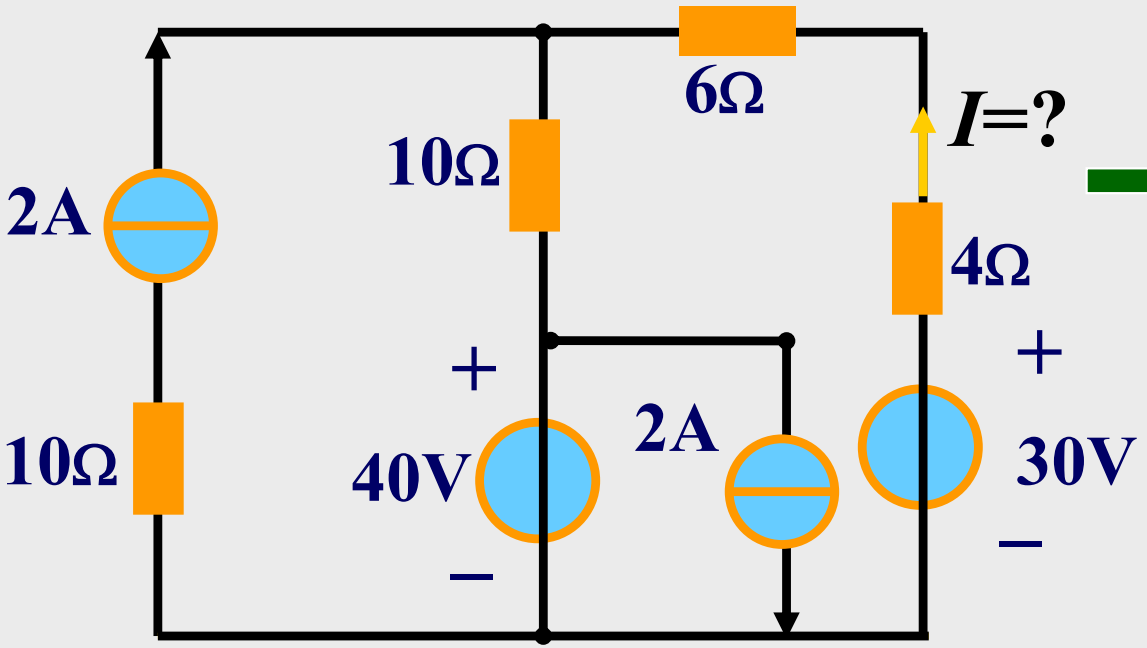
**例:** 试用电压源与电流源等效变换的方法计算2Ω电阻中的电流。

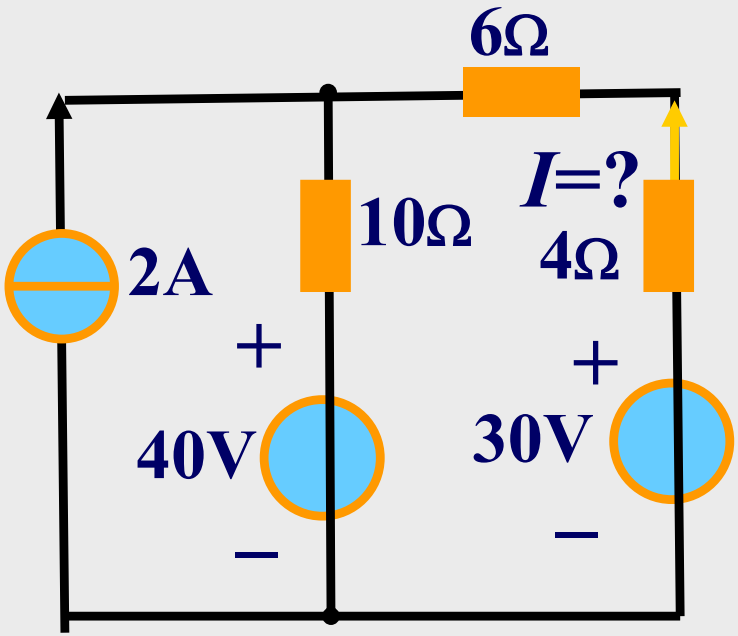


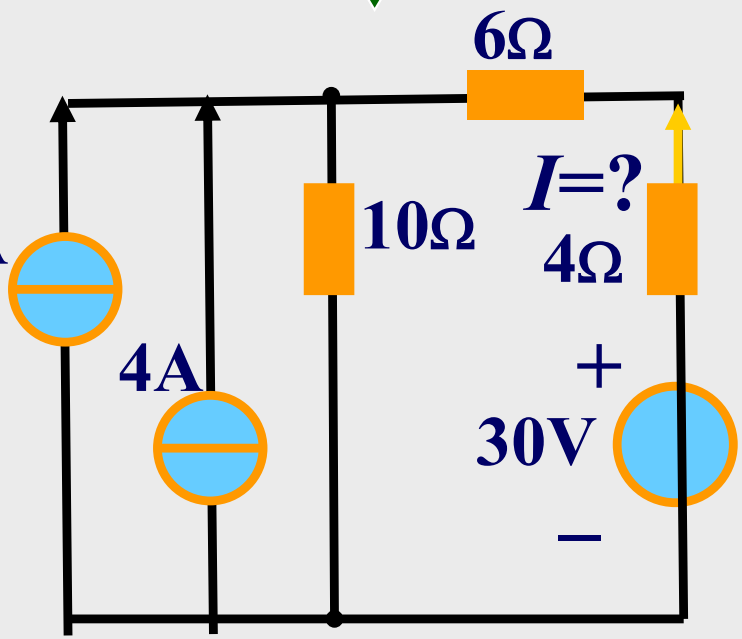

由图(d)可得

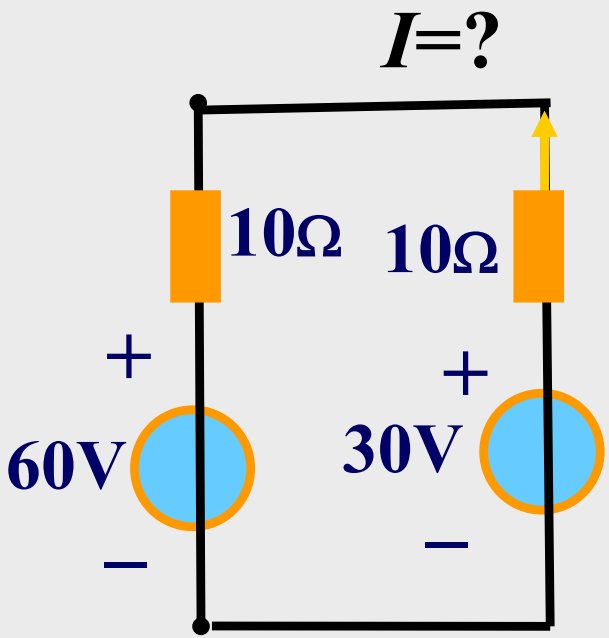
$$I = \frac{8-2}{2+2+2}\,\mathrm{A} = 1\,\mathrm{A}$$

例.



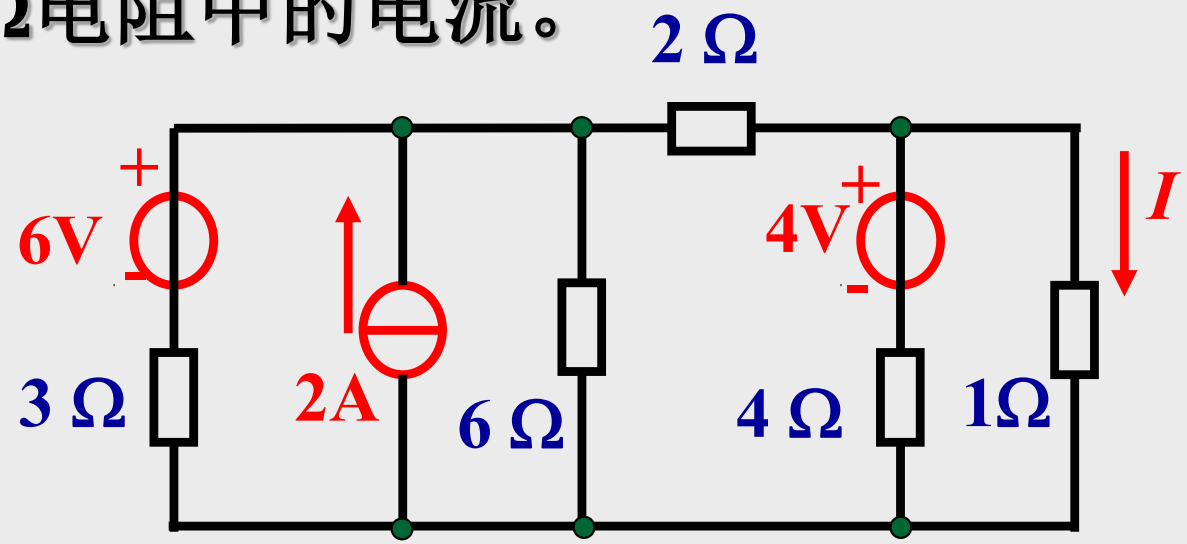



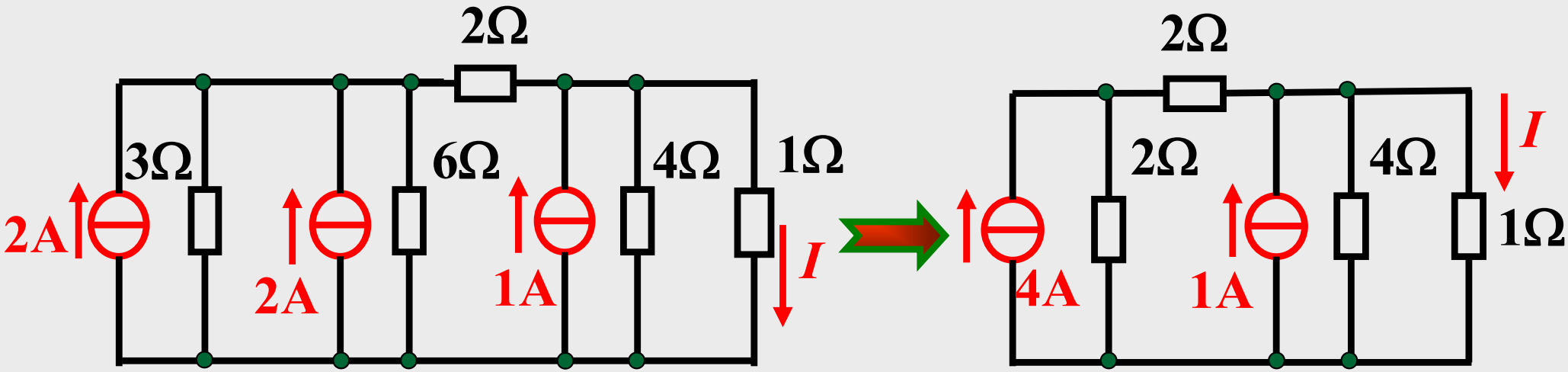




$$I = \frac{30 - 60}{20} = -1.5A$$

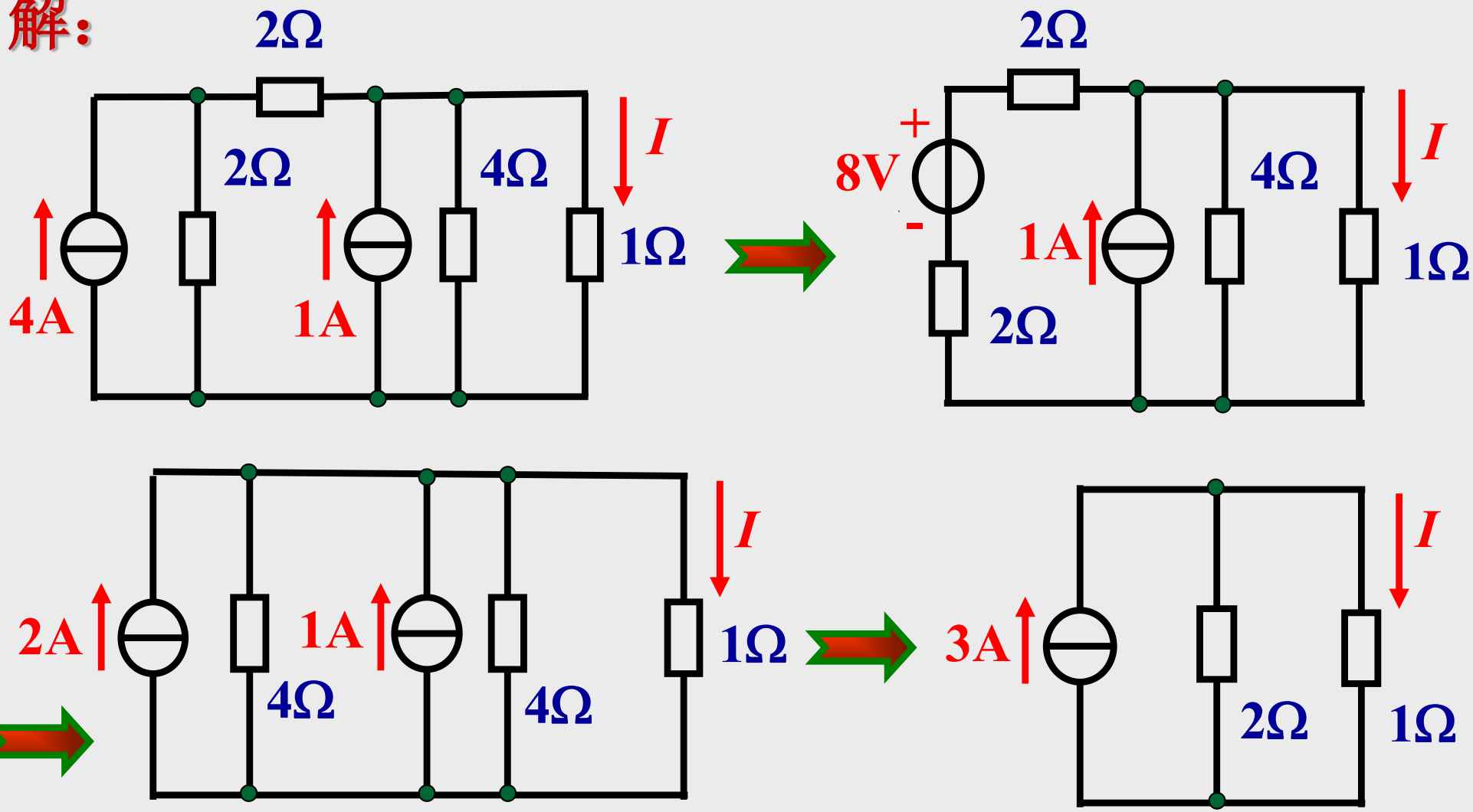
例：试用电压源与电流源等效变换的方法计算图示电路中1 Ω电阻中的电流。




解：统一电源形式

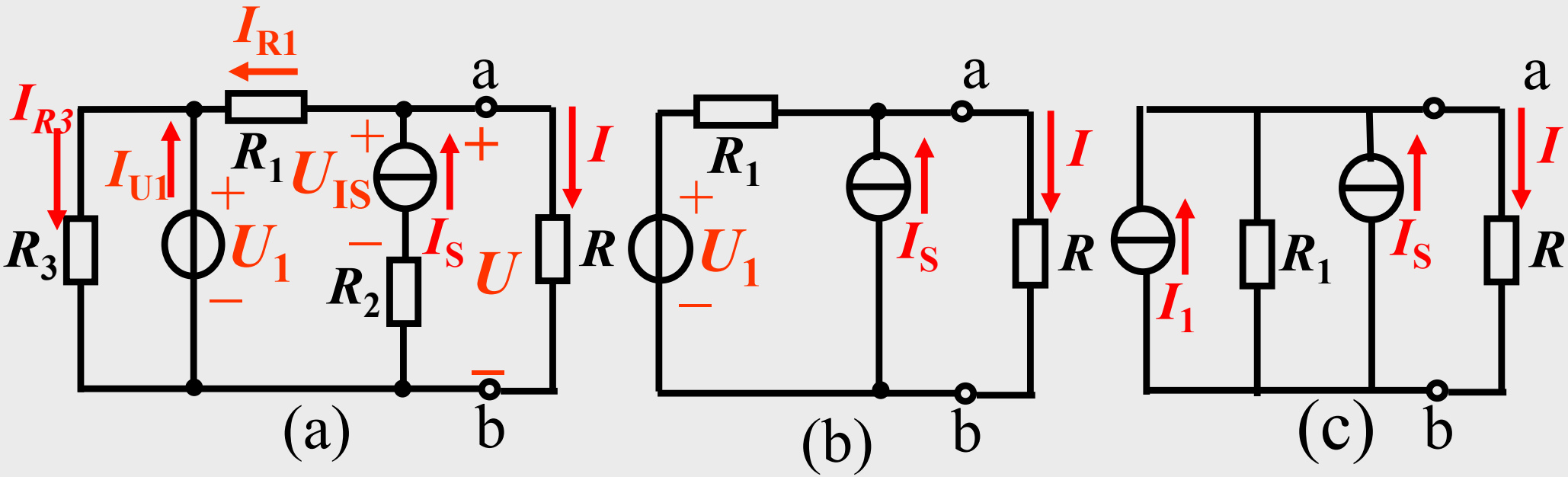


解：



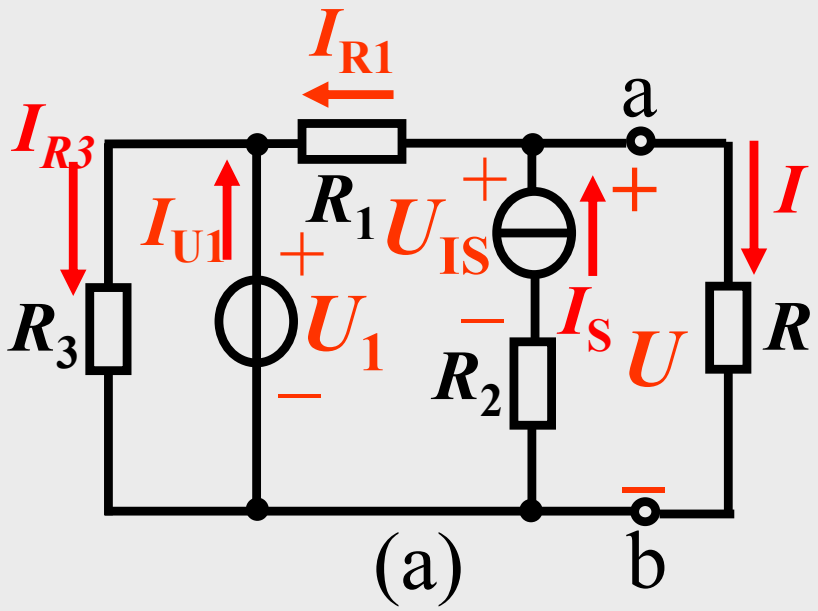

$$I = \frac{2}{2+1} \times 3\,\mathrm{A} = 2\,\mathrm{A}$$

**例：** 电路如图。$U_1=10\text{V}$，$I_S=2\text{A}$，$R_1=1\Omega$，$R_2=2\Omega$，$R_3=5\,\Omega$，$R=1\,\Omega$。(1) 求电阻$R$中的电流$I$；(2)计算理想电压源$U_1$中的电流$I_{U1}$和理想电流源$I_S$两端的电压$U_{IS}$；(3)分析功率平衡。




**解：(1)由电源的性质及电源的等效变换可得：**

$$I_1=\frac{U_1}{R_1}=\frac{10}{1}\text{A}=10\text{A}\qquad I=\frac{I_1+I_S}{2}=\frac{10+2}{2}\text{A}=6\text{A}$$




(2)由图(a)可得：

$$I_{R1} = I_S - I = 2\text{A} - 4\text{A} = -4\text{A}$$

$$I_{R3} = \frac{U_1}{R_3} = \frac{10}{5}\text{A} = 2\text{A}$$

理想电压源中的电流

$$I_{U1} = I_{R3} - I_{R1} = 2\text{A} - (-4)\text{A} = 6\text{A}$$

理想电流源两端的电压

$$U_{IS} = U + R_2 I_S = RI + R_2 I_S = 1\times 6\text{V} + 2\times 2\text{V} = 10\text{V}$$

(3)由计算可知，本例中理想电压源与理想电流源都是电源，吸收的功率分别是：

$$P_{U1} = -U_1 I_{U1} = -10 \times 6 = -60\text{W}$$

$$P_{IS} = -U_{IS} I_S = -10 \times 2 = -20\text{W}$$

各个电阻所消耗的功率分别是：

$$P_R = RI^2 = 1 \times 6^2 = 36\text{W}$$

$$P_{R1} = R_1 I_{R1}^2 = 1 \times (-4)^2 = 16\text{W}$$

$$P_{R2} = R_2 I_S^{\ 2} = 2 \times 2^2 = 8\text{W}$$

$$P_{R3} = R_3 I_{R3}^{\ 2} = 5 \times 2^2 = 20\text{W}$$

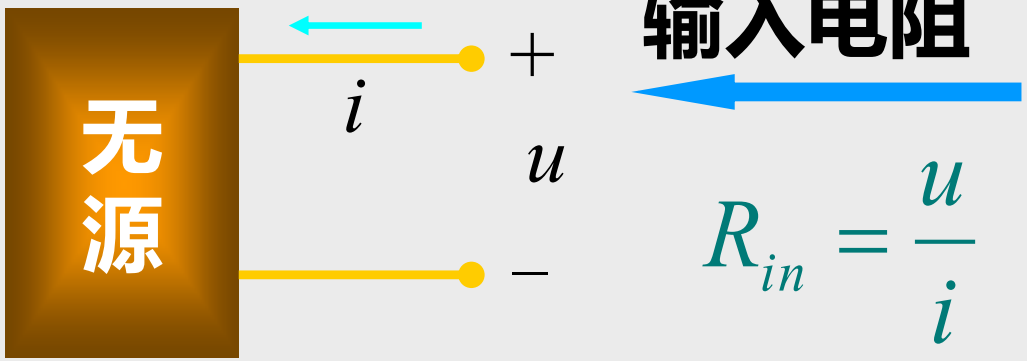
两者平衡：

$$(60+20)\text{W} = (36+16+8+20)\text{W}$$

$$80\text{W} = 80\text{W}$$

# 2.6 输入电阻

## 1.定义




$$R_{in}=\frac{u}{i}$$

## 2.计算方法

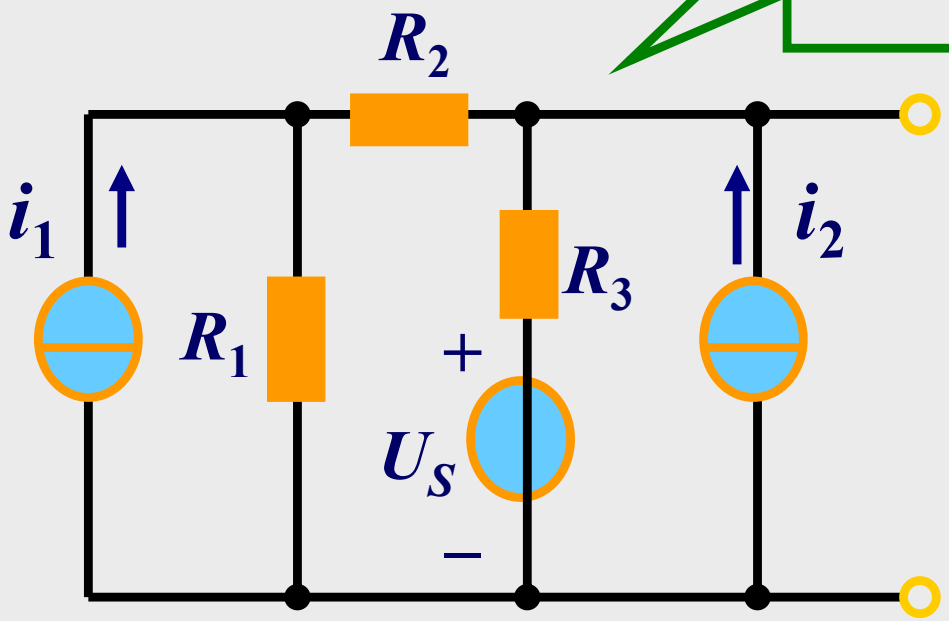
①如果一端口内部仅含电阻，则应用电阻的串、并联和Δ—Y变换等方法求它的等效电阻；

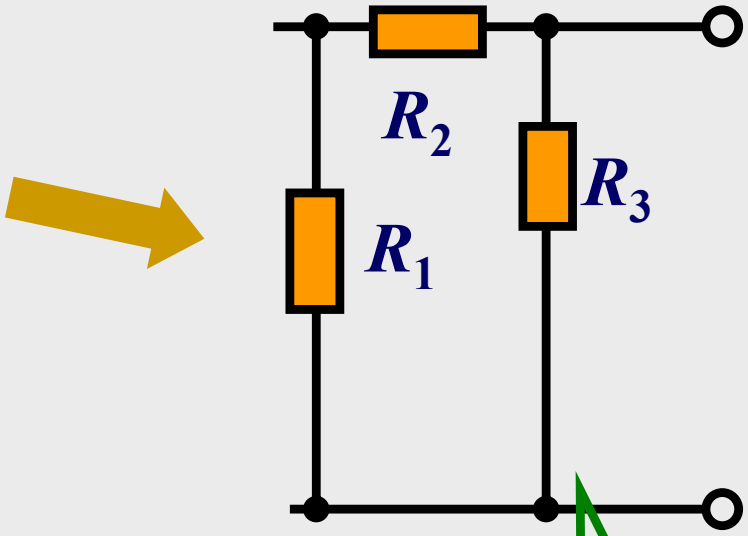
②对含有受控源和电阻的两端电路，用电压、电流法求输入电阻，即在端口加电压源，求得电流，或在端口加电流源，求得电压，得其比值。

# 计算下例一端口电路的输入电阻

**例1.**

有源网络先把独立源置零：电压源短路；电流源断路，再求输入电阻

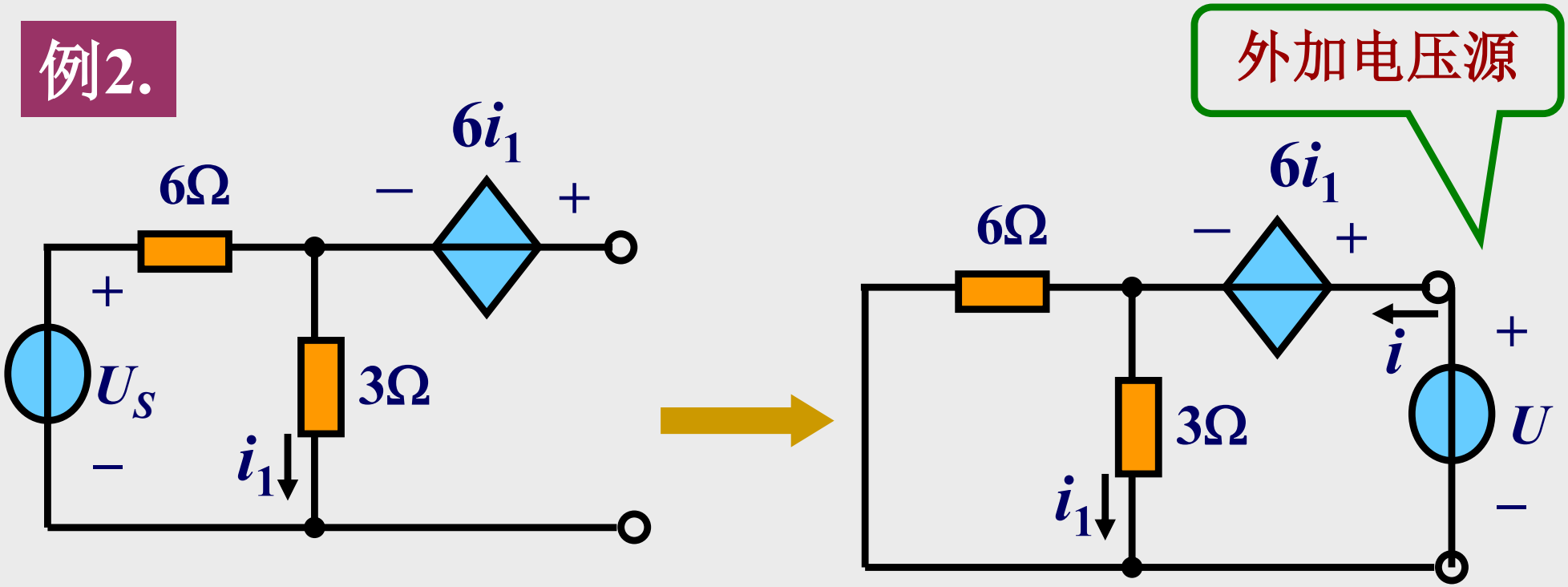



无源电阻网络

$$R_{in} = (R_1 + R_2) // R_3$$

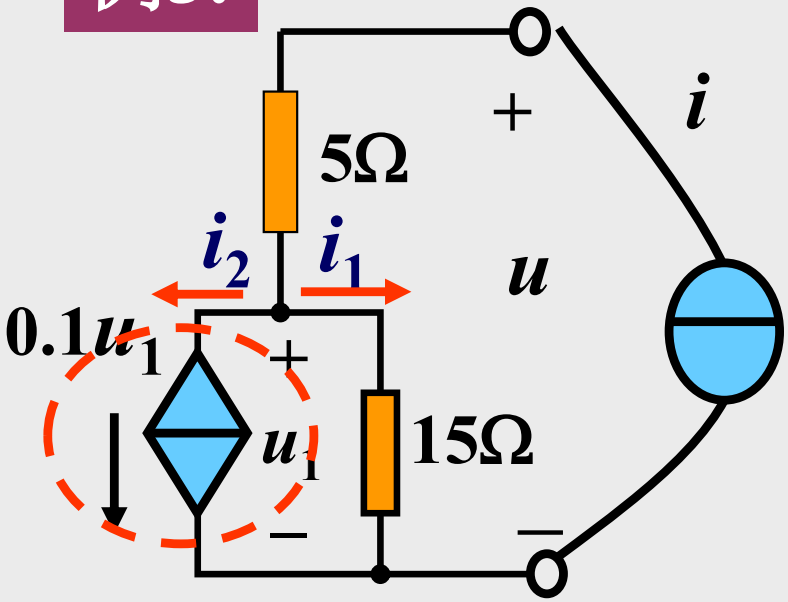
**例2.**



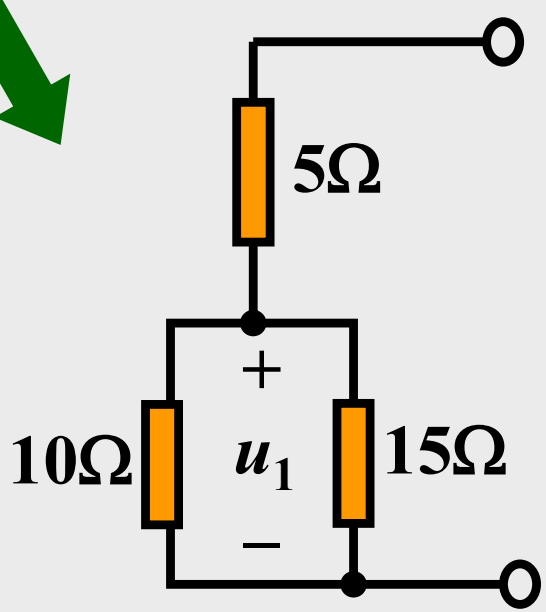

$$i = i_1 + \frac{3i_1}{6} = 1.5i_1 \qquad U = 6i_1 + 3i_1 = 9i_1$$

$$R_{in} = \frac{U}{i} = \frac{9i_1}{1.5i_1} = 6\Omega$$

**例3.**




等效




$$u_1 = 15i_1 \quad i_2 = \frac{u_1}{10} = 1.5i_1$$

$$i = i_1 + i_2 = 2.5i_1$$

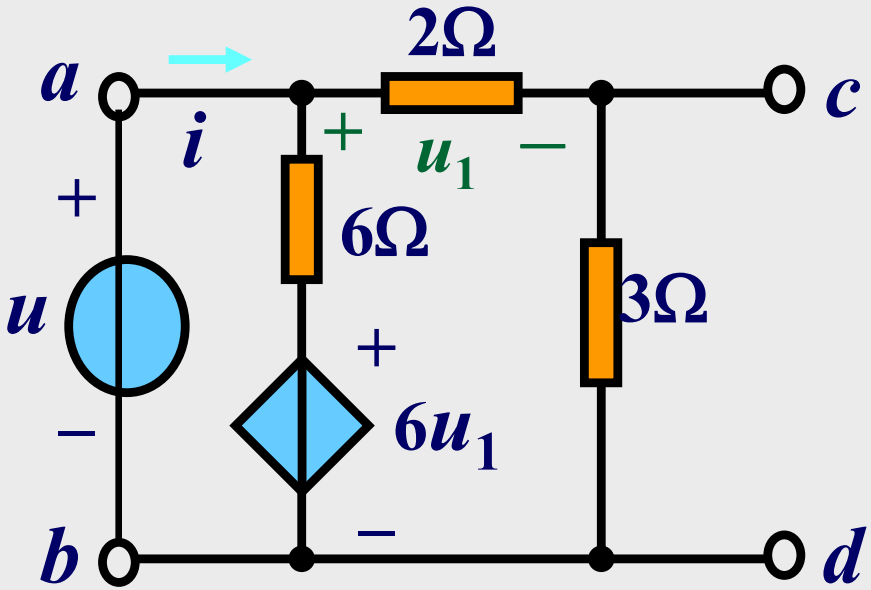
$$u = 5i + u_1 = 5 \times 2.5i_1 + 15i_1$$

$$= 27.5i_1$$

$$R_{in} = \frac{u}{i} = \frac{27.5i_1}{2.5i_1} = 11\Omega$$

$$R_{in} = 5 + \frac{10 \times 15}{10 + 15} = 11\Omega$$

**例.** 求$R_{ab}$



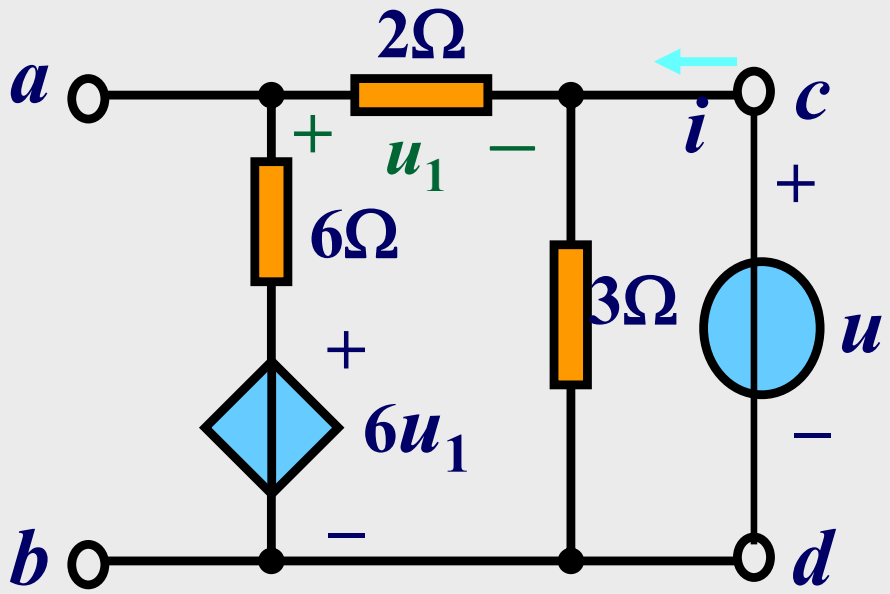

$$u_{ab} = u_1 + 3u_1/2 = 2.5u_1$$

$$u_1 = u_{ab}/2.5 = 0.4\,u_{ab}$$

$$i = \frac{u_1}{2} + \frac{u_{ab} - 6u_1}{6} = -u_{ab}/30$$

$$R_{ab} = u_{ab}/i = -30\Omega$$

**例.** 求$R_{cd}$

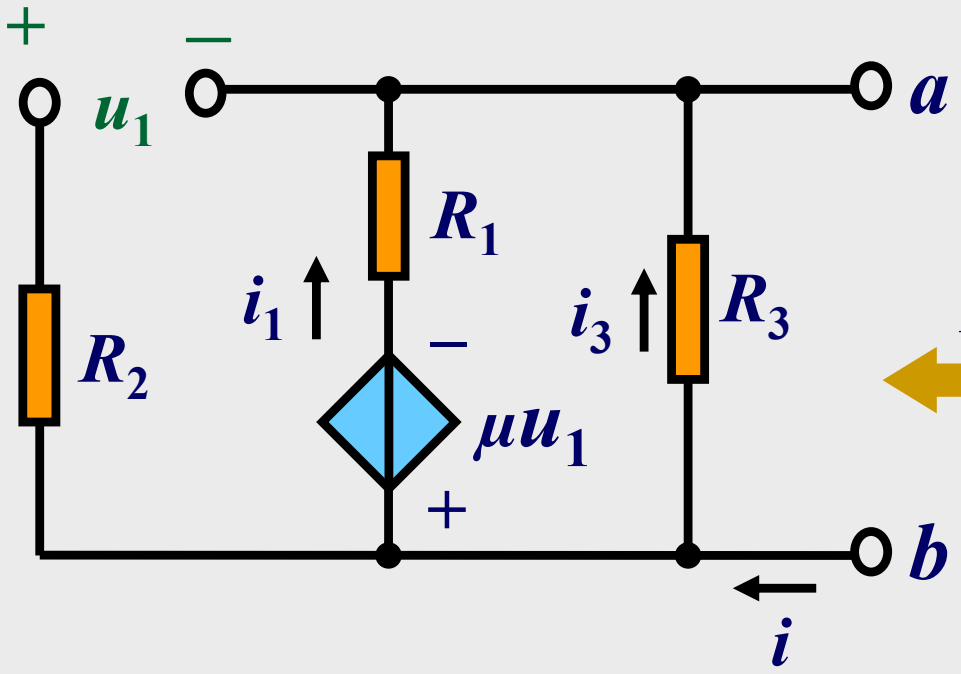



$$u_{cd} = -u_1 + 6u_1 + \frac{6\times(-u_1)}{2} = 2u_1$$

$$i = -u_1/2 + u_{cd}/3 = u_1/6$$

$$R_{cd} = u_{cd}/i = 12\Omega$$

例.

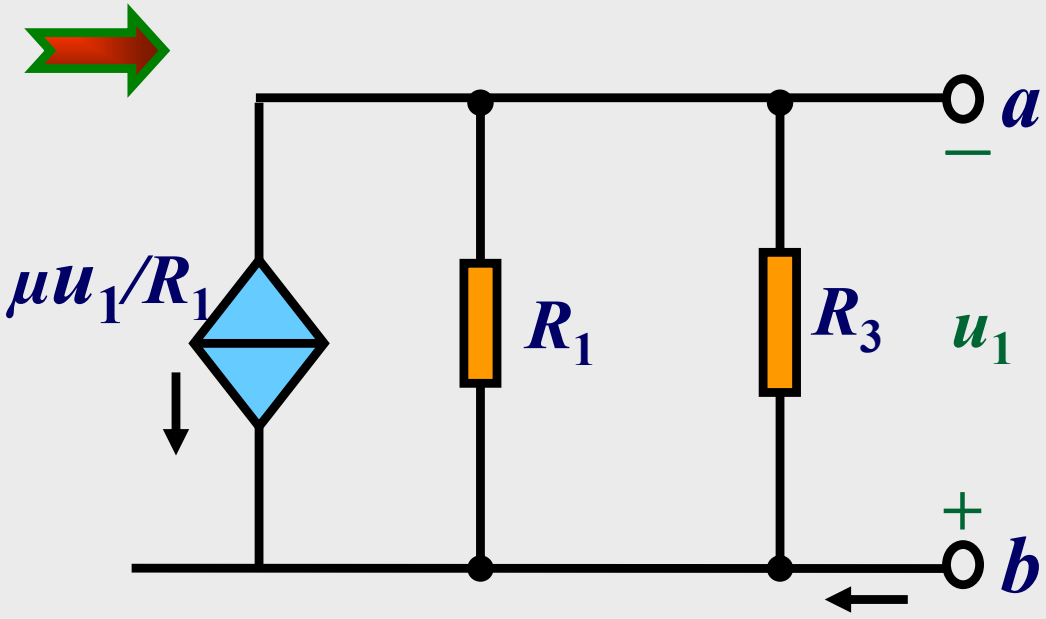



$R_i$

$u_1 = i_3 R_3 = \mu u_1 + i_1 R_1$

$i_3 + i_1 = i$

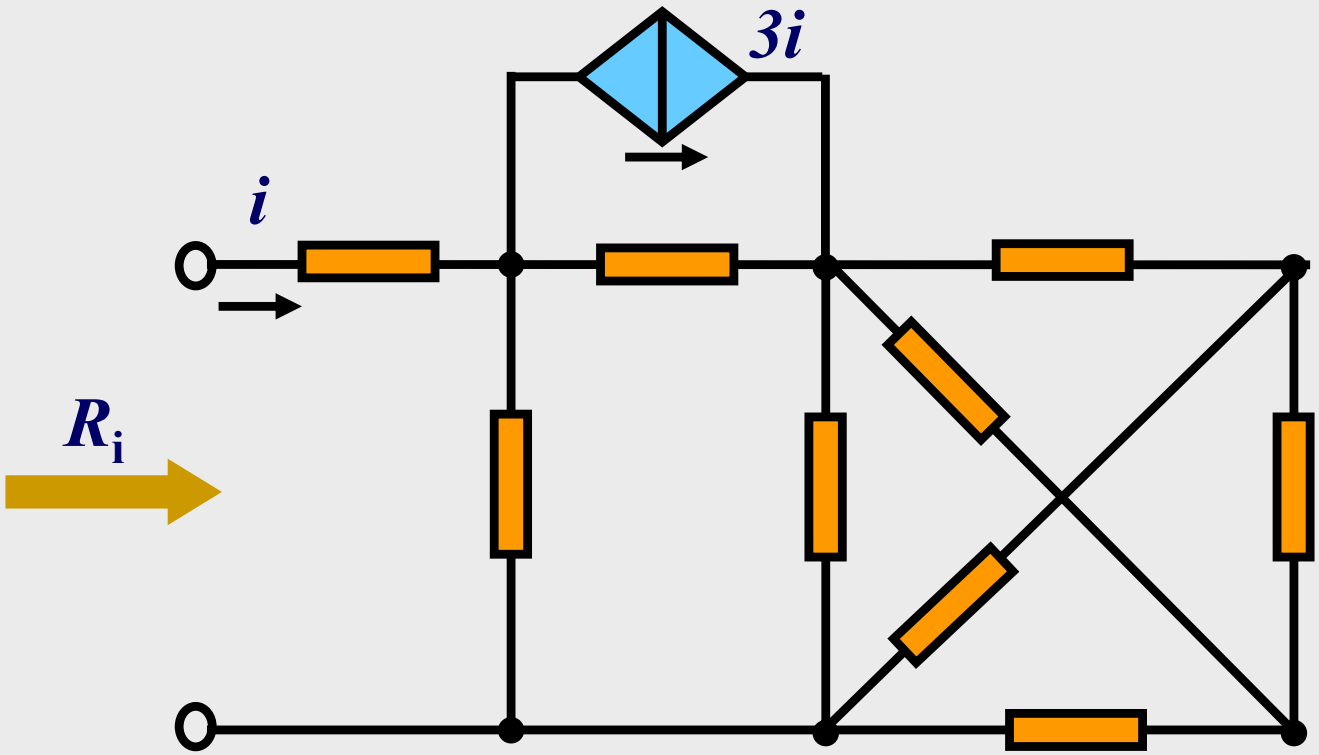
$R_i = \dfrac{u_i}{i}$

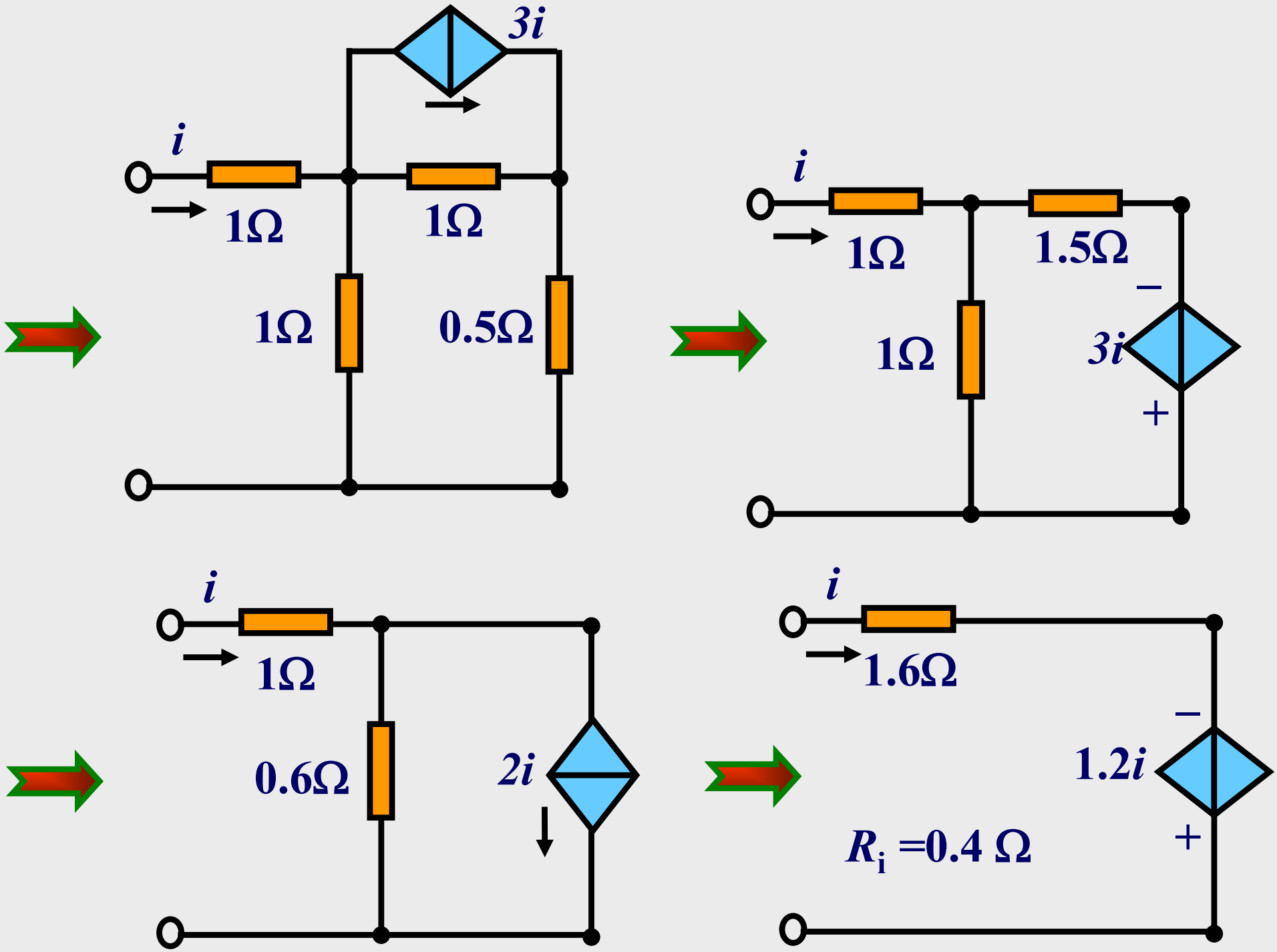


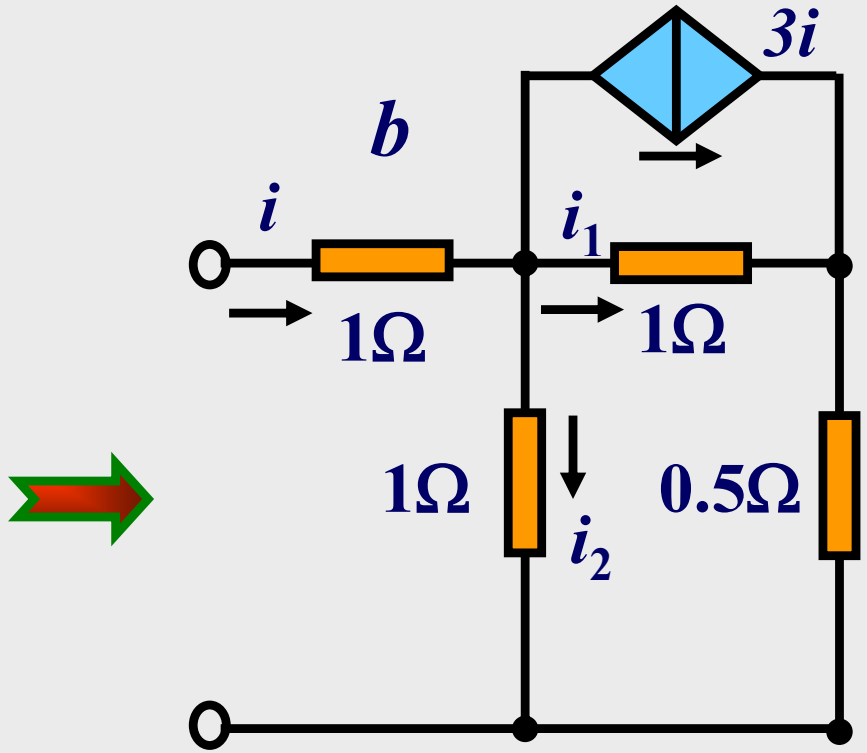

$R_i = (-R_1/\mu) // R_1 // R_3$

例.

所有电阻都是R，R=1 Ω




3i
i
1Ω
1Ω
1Ω
0.5Ω
i
1Ω
1.5Ω
1Ω
3i
i
1Ω
0.6Ω
2i
i
1.6Ω
1.2i
$R_i$ =0.4 Ω




$$3i + i_1 + i_2 = i$$

$$i_2 = i_1 + 0.5 * (3i + i_1)$$

$$i_2 = -0.6i$$

$$u = i + i_2 = 0.4i$$

$$R_i = 0.4\ \Omega$$

# 第二章 总结

1.电阻的Y形连接和△形连接的等效变换；

2.实际电源的两种模型及其等效变换；

3.输入电阻的计算方法。

**作业**：

P46-P51：

2-1；2-4(c)(d)(f)；2-7；

2-11；2-14(b)；

五题任意选做三题，下周五下课前提交作业

The End